

## Research papers

# A novel state of health estimation method for lithium-ion battery based on multi-feature extraction and fusion model with bidirectional feature extraction

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## ABSTRACT

Accurate estimation of the state of health (SOH) of lithium-ion batteries is essential for effective battery system management and maintenance. Conventional data-driven methods often fail to adequately utilize bidirectional temporal correlations in degradation trajectories and capture long-term aging dependencies. This paper proposes a novel method based on a BiTCN-BiGRU-MHSA fusion model. Initially, health factors (HFs) correlated with capacity degradation are extracted from charge–discharge voltage and current curves. Subsequently, HF outliers are identified using the local outlier factor (LOF) algorithm, replaced by Lagrange interpolation, and denoised via singular spectrum analysis (SSA). To validate the HFs, Pearson and Spearman correlation coefficients are combined for screening. Features with strong SOH correlation are selected for the dataset. The core approach involves: (1) Bidirectional features are extracted by bidirectional temporal convolutional network (BiTCN) to address traditional one-way bias. (2) Features are deeply fused with bidirectional gated recurrent unit (BiGRU) for long-term dependency modeling. (3) Focus on critical patterns is enhanced via multi-head self-attention (MHSA), strengthening attention to degradation-sensitive features. The BiTCN-BiGRU-MHSA fusion model is formed by this integrated framework, establishing a multi-level feature extraction system. Finally, the method is validated on NASA, CALCE, and XJTU datasets. The experimental results show a significant reduction in mean absolute error (MAE) versus models like CNN-CBAM-LSTM and Informer. Specifically, on the NASA dataset, the average MAE is reduced by 30% and 39%, respectively. On the CALCE dataset, reductions of 23% and 31% are achieved. On the XJTU dataset, reductions of 10% and 14% are achieved.

## 1. Introduction

The lithium-ion battery represents an advanced high-energy-density storage device, characterized by significant advantages including extended cycle life and absence of memory effect. Owing to these attributes, lithium-ion batteries have achieved widespread deployment in critical applications such as new energy vehicles and aerospace systems [1]. However, battery performance inevitably degrades over time. With increasing operational cycles and prolonged storage, lithium-ion batteries undergo progressive aging, manifesting as declines in both power capability and capacity [2,3]. The state of health (SOH) quantifies the degradation process of batteries. SOH is predominantly defined by either internal resistance or capacity — with capacity serving as the primary metric — and it is the critical indicator of battery aging [4,5]. Consequently, accurate SOH estimation is essential, as it facilitates the identification of aged batteries and provides vital support for effective battery management and maintenance strategies.

The SOH estimation methods for lithium-ion batteries are predominantly categorized into model-based and data-driven approaches [6,7]. The model-based approach primarily encompasses empirical degradation models (EDM), equivalent circuit models (ECM), and electrochemical models (EM) [8]. An EDM-based approach was proposed by Tseng et al. [9], where full discharge voltage ( $V_{dis}$ ) and internal resistance ( $R$ ) were employed as aging parameters. Polynomial and exponential regression models were formulated, with coefficients optimized via particle swarm optimization (PSO). This approach achieved a coefficient of determination of 98% for the SOH estimation. An ECM-based technique was developed by Wei et al. [10], which utilized a first-order Thevenin model to capture the strong capacitive characteristics of batteries. In this technique, Health indices (HI) were defined by integrating the increases in internal or polarization resistance with the decreases in polarization capacitance. Additionally, offline training

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of the model via gray neural networks maintained the estimation errors below 2%. Zhang et al. introduced an EIS-ECM-probabilistic fusion method [11], where the impacts of temperature and state-of-charge (SOC) on electrochemical impedance spectroscopy (EIS) spectra were quantified. Dynamic-condition estimation errors were reduced to approximately 4%, outperforming conventional EIS-based approaches. In parallel research, the impedance characteristics of the pseudo-2-dimensional (P2D) model were analyzed across different frequency domains [12]. It was observed that impedance alignment with physical batteries occurs exclusively in low-frequency ranges. To enhance the applicability of the P2D model, a modified model — incorporating low-frequency P2D features — was developed and validated against the standard P2D model, achieving a 55.7% reduction in mean absolute error (MAE). Despite advantages in data efficiency and accuracy, the model-based approach faces constraints, including physical parameter identification challenges and implementation limitations.

In contrast to the model-based approach, the data-driven approach abstracts from internal electrochemical reactions and failure mechanisms, circumventing the need for complex process analysis. Historical operational data are leveraged to extract capacity-degradation-related features, capture battery aging characteristics, and establish the SOH estimation models [13,14]. Currently, practitioners mainly employ machine learning and deep learning approaches. For example, Li et al. developed an SOH estimation method called IALO-SVR [15], which leveraged an improved ant-lion optimization algorithm (IALO) in conjunction with support vector regression (SVR). This method extracted health features closely linked to the SOH degradation as inputs for the SVR model while using the IALO algorithm to tune the kernel parameters during training. The experimental results showed that the SOH estimation error remained stable at 2% or less. Chen et al. introduced an ELM-based method for the SOH estimation [16]. Features related to battery capacity degradation were first extracted from battery charging and incremental capacity curves. Then, the correlation between these features and battery capacity was evaluated using gray correlation analysis (GRA). The refined features served as inputs to the ELM, with parameters tuned using the PSO algorithm. The experimental results demonstrated that the method achieved a mean absolute percentage error (MAPE) of less than 1% in the SOH estimation. Toughzaoui et al. proposed an SOH estimation method that integrates a convolutional neural network (CNN), a long short-term memory network (LSTM), and a K-means clustering algorithm [17]. K-means clustering was implemented to classify data, thereby optimizing the feature extraction process effectively. Then, features extracted via the CNN were input into the LSTM for the SOH estimation in a CNN-LSTM hybrid model, achieving a root mean square error (RMSE) of 1.4%. Zhang et al. proposed an SOH estimation method utilizing the LSTM [18], constructing a model that took historical capacity data of lithium-ion batteries as input. The experimental results indicated that the RMSE of the SOH estimation was less than 5%. Ouyang et al. introduced a method that combines model-agnostic meta-learning (MAML) with CNN-LSTM [19]. The CNN-LSTM was first pre-trained using a meta-learning framework to capture generic battery features, then fine-tuned with specific battery data to adapt to various battery types. The experimental results showed that this approach performed well across multiple battery datasets, achieving a maximum RMSE of 2.31% and an MAE of 2.03%. Yu et al. developed an SOH estimation method combining incremental expansion analysis (IEA) and CNN-LSTM-Attention [20], which leveraged the IEA method in conjunction with the CNN-LSTM-Attention model. This method generated the incremental expansion (IE) curve by tracking the expansion increment associated with voltage steps, analyzed the IE curve under different working conditions to identify its regularity and correlation with battery aging, and then extracted peak features closely linked to the SOH degradation as health indicators for the CNN-LSTM-Attention model. The experimental results demonstrated that this method achieved high estimation accuracy and robustness, with the MAE within 0.12%. Xu et al. developed an SOH estimation method for

lithium-ion batteries based on incremental capacity analysis (ICA) and Transformer [21]. This method first extracted the original incremental capacity (IC) curve of the battery using the ICA method, then processed the original IC curve with a dual filtering method combining a moving average smoothing filter (MASF) and a Gaussian smoothing filter (GSF) to reduce noise, and further extracted peak features of the IC curve. After that, a Transformer network model based on the multi-head attention mechanism was constructed, with the extracted IC curve peak features as model inputs to realize the SOH estimation of lithium-ion batteries. The experimental results showed that the proposed SOH estimation method has higher long-term prediction accuracy and better temperature adaptability than commonly used machine learning methods.

While existing methods enhance the SOH estimation accuracy to a certain degree, they exhibit two limitations: inadequate exploitation of bidirectional temporal correlations in battery degradation trajectories and insufficient capture of long-term dependencies. Conventional unidirectional architectures, such as LSTM and CNN-LSTM, neglect the influence of future degradation states on current states, while shallow network structures lack the capacity to model extended degradation patterns. To address these issues, this paper proposes a novel SOH estimation method based on a BiTCN-BiGRU-MHSA fusion model. The method initiates with extracting health factors (HFs) from lithium-ion battery charge–discharge curves, which characterize degradation mechanisms. Subsequently, the HFs undergo preprocessing through local outlier factor (LOF) detection and Lagrange interpolation for outlier mitigation, followed by singular spectrum analysis (SSA) denoising to enhance signal quality. Feature validation employs Pearson and Spearman correlation analysis methods to verify HF significance. Highly correlated features are selected as model inputs, constituting the experimental dataset. The BiTCN-BiGRU-MHSA fusion model architecture integrates BiTCN with BiGRU, augmented by an MHSA mechanism. Unlike conventional approaches, the bidirectional temporal convolutional network (BiTCN) extracts multi-scale temporal features capturing bidirectional degradation correlations: past-to-current state relationships and future-to-current state relationships. These fused features are propagated to the bidirectional gated recurrent unit (BiGRU) for comprehensive temporal dependency modeling. The multi-head self-attention (MHSA) mechanism subsequently applies adaptive weighting to emphasize critical degradation indicators. Final SOH estimation is generated through a fully connected output layer. To ensure robust learning of HF-capacity degradation relationships, experimental validation reserves one battery sample each from NASA, CALCE, and XJTU datasets for independent testing, utilizing the remaining samples for training and validation. Model optimization minimizes mean square error (MSE) loss during hyperparameter tuning. Generalization capability is ultimately assessed on the test set prior to deployment for battery SOH estimation.

The contributions of this paper are as follows:

- (1) Capacity-related HFs are extracted from Li-ion battery charge–discharge curves, with outliers identified by LOF and replaced by Lagrange interpolation, thereby preserving data integrity and reducing subjective errors.
- (2) SSA is used to denoise the interpolated HFs to reduce noise interference, thus further improving the data quality.
- (3) HFs are selected via dual correlation coefficients to build a reliable dataset.
- (4) A BiTCN-BiGRU-MHSA model is proposed, solving traditional bidirectional and long-term dependency issues while providing strong feature extraction capabilities.

## 2. Methodology

Conventional lithium-ion battery SOH estimation methods face two challenges: poor characterization of the correlation between future and

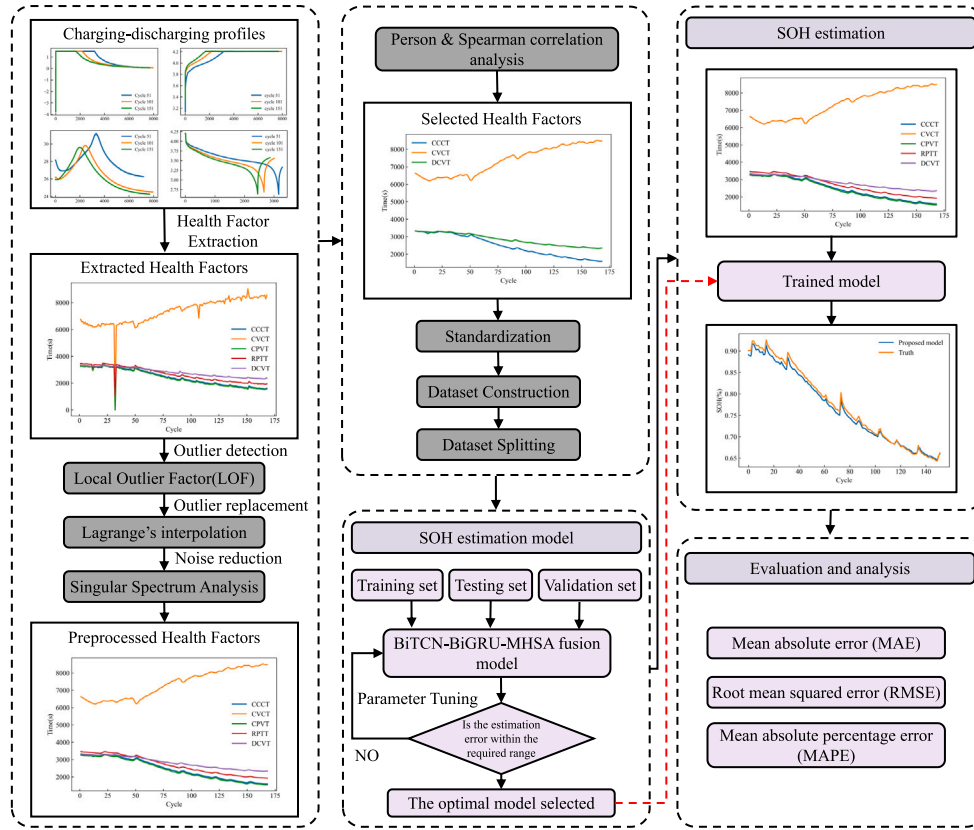


Fig. 1. The framework for estimating the SOH of lithium-ion batteries.

current states during degradation and limited ability to capture long-term temporal dependencies. To overcome these issues, a novel SOH estimation method based on a BiTCN-BiGRU-MHSA fusion model is proposed. The overall framework of the proposed method, as shown in Fig. 1, includes four key components: data preprocessing, dataset construction, the SOH estimation model, and experimental design with result analysis.

**Data preprocessing:** HF's related to battery capacity degradation are extracted from the charge-discharge curves of lithium-ion batteries. Subsequently, the LOF algorithm is employed to detect outliers in the obtained HF's, and Lagrange interpolation is used to replace the outliers. Finally, SSA is applied to denoise the extracted HF's.

**Dataset construction:** The relationship between preprocessed HF's and the SOH of lithium-ion battery is first assessed using the Pearson correlation coefficient and the Spearman correlation coefficient. Subsequently, the most relevant HF's are selected as input features to construct the experimental dataset. Next, the dataset is standardized to minimize the impact of dimensional discrepancies across features on model convergence. Finally, the dataset is partitioned into a training set, validation set, and test set, which are respectively utilized in the training, validation, and testing of model experiments.

**SOH estimation model:** To enhance the extraction of bidirectional correlations and long-term temporal dependencies, a BiTCN is integrated with a BiGRU, with an MHSA mechanism incorporated. This integration results in a BiTCN-BiGRU-MHSA fusion model designed to extract bidirectional temporal features and capture long-term dependencies. Bidirectional local features are first extracted by BiTCN's dilated causal convolutions, which are then processed by BiGRU to model long-term dependencies across time steps. Key temporal features are adaptively emphasized through the MHSA mechanism's dynamic weighting strategy, enabling the creation of hierarchical feature representations. Subsequently, during the training phase, the training and validation sets are utilized to train the model. Finally, in the testing

phase, the test set is employed to evaluate the model's generalization capability across different battery types, verifying the effectiveness of the fusion model.

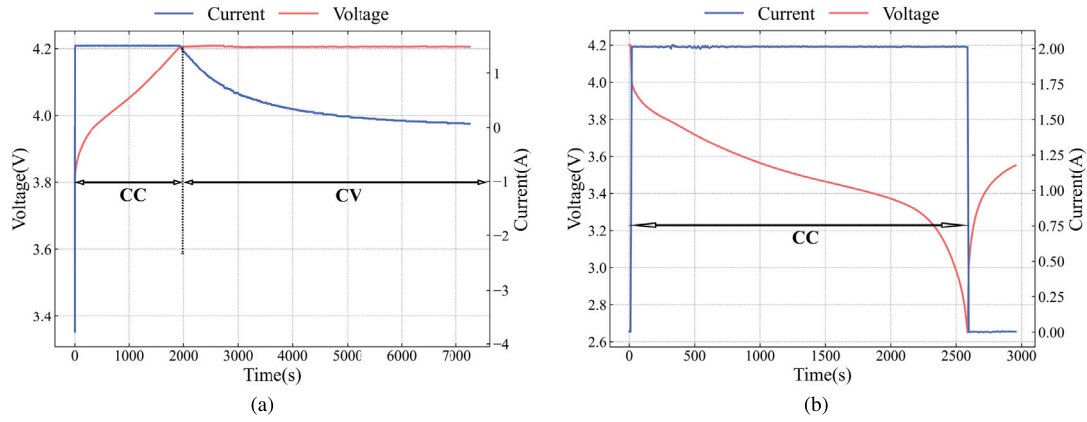
**Experimental design with result analysis:** The proposed method is evaluated against existing methods using metrics of RMSE, MAE, and MAPE, with subsequent experimental results subjected to comprehensive statistical analysis.

## 2.1. Data preprocessing

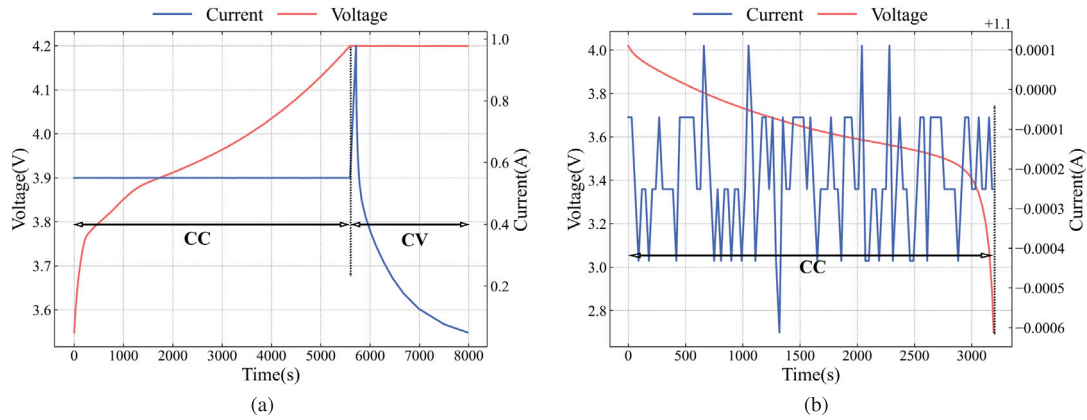
### 2.1.1. Battery dataset

The three lithium-ion battery datasets used in this paper are obtained from the National Aeronautics and Space Administration (NASA), the Center for Advanced Life Cycle Engineering (CALCE) at the University of Maryland [22], and Xi'an Jiao Tong University (XJTU) [23]. The NASA lithium battery dataset includes lithium batteries numbered B0005, B0006, B0007, and B0018, while the CALCE lithium battery dataset comprises lithium batteries numbered CS2\_35, CS2\_36, CS2\_37, and CS2\_38. For the XJTU lithium battery dataset, Batteries 4, 5, and 6 from Batch 4 are selected in this paper. The detailed information on the charge-discharge experiments of lithium-ion batteries in the three datasets is shown in Table 1.

The experimental procedures of the NASA, CALCE, and XJTU lithium battery datasets involve repeated charge-discharge cycles. The charging process consists of two stages: a constant current (CC) charging stage and a constant voltage (CV) charging stage, while the discharging process involves a CC discharging stage. As illustrated in Fig. 2, the charge-discharge process of NASA lithium batteries is as follows: during charging, a CC mode of 1.5 A is first applied until the voltage reaches 4.2 V, followed by a CV mode until the charging current drops to 20 mA. The discharging process is conducted at a CC of 2 A until the voltage reaches the cut-off voltage.



**Fig. 2.** Current and voltage curves during charging and discharging of lithium-ion battery B0005. (a) Current and voltage curves during charging. (b) Current and voltage curves during discharging.



**Fig. 3.** Current and voltage curves during charging and discharging of lithium-ion battery CS2\_35. (a) Current and voltage curves during charging. (b) Current and voltage curves during discharging.

For the CALCE lithium batteries, as shown in Fig. 3, the charge–discharge process is: during charging, a CC of 0.55 A is initially used until the voltage reaches 4.2 V, then is switched to CV mode until the charging current drops to 50 mA. The discharging process is performed at a CC of 1.1 A until the voltage drops to the cut-off voltage.

For the XJTU lithium batteries, to measure the initial capacity of the battery in the first cycle: a CC of 1 A is adopted to charge until the voltage reaches 4.2 V, then the voltage is maintained constant until the current drops to 40 mA; the battery is discharged at a CC of 0.4 A until the voltage reaches 2.5 V. For other cycles: the batteries are charged at a CC of 4 A until the voltage reaches 4.2 V, then the voltage is maintained constant until the current drops to 0.1 A; the batteries are discharged at  $x$  A until the voltage reaches 3.0 V, where  $x$  cycles through the set {1,2,4,6,10}; whenever  $x$  completes one round of cycling, the following operation is performed to measure the capacity: the battery is charged at a CC and CV of 4 A until the voltage reaches 4.2 V; the battery is discharged at 2 A until the voltage reaches 2.5 V. Fig. 4 shows the charge–discharge process in the third cycle.

With the progression of charge–discharge cycles in lithium-ion batteries, the aging phenomenon is inevitable, and the most significant change in external performance is manifested as battery capacity degradation [24]. Specifically, the electrolyte and active materials of the positive and negative electrodes continuously consume lithium ions, leading to an increase in charge transfer impedance on the battery surface and capacity loss. The capacity degradation curves of NASA, CALCE, and XJTU lithium batteries are shown in Figs. 5(a), 6(a), and 7(a), respectively. However, under certain specific conditions, the

battery capacity may suddenly increase after a charge–discharge cycle, a phenomenon known as capacity regeneration. This is an inherent degradation characteristic of lithium batteries rather than a measurement error. The SOH serves as a key indicator for characterizing battery aging [25]. Although the SOH does not have a universally standardized definition, it is typically defined as the ratio of the current capacity to the rated capacity, calculated as shown in Eq. (1). Figs. 5(b), 6(b), and 7(b) present the SOH degradation curves of NASA, CALCE, and XJTU.

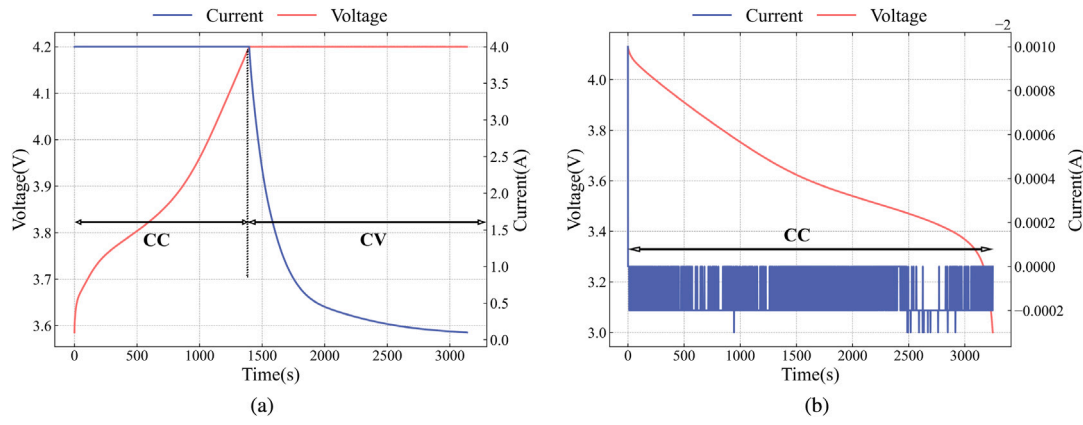
$$SOH = \frac{C_P}{C_N} \times 100\% \quad (1)$$

where  $C_P$  denotes the current capacity and  $C_N$  denotes the rated capacity.

### 2.1.2. Health factor extraction

During battery aging, capacity directly indicates the degree of battery aging, but it is not easily measurable. Conversely, HFs extracted from voltage, current, and temperature profiles that correlate with battery capacity degradation can indirectly reflect battery aging. Therefore, this paper extracts HFs related to battery capacity degradation from the charge–discharge curves of NASA, CALCE, and XJTU lithium batteries. Constant current charge time (CCCT) refers to the duration of the constant current charging stage. Fig. 8(a) shows the current profile of B0005 during charge–discharge cycles, where CCCT decreases with increasing cycle count. For example, the CCCT in cycle 42 is shorter than that in cycle 82, a trend observed in other cycles. Mechanistically, with repeated charge–discharge cycles, SEI film thickening and active

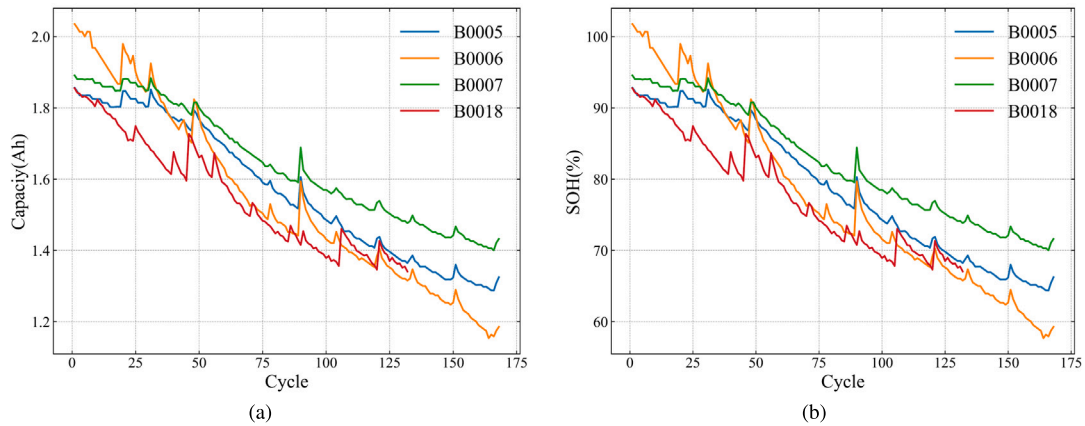




**Fig. 4.** Current and voltage curves during charging and discharging of lithium-ion battery Battery-4. (a) Current and voltage curves during charging. (b) Current and voltage curves during discharging (2 A).

**Table 1**  
Battery experiment information.

| Battery   | Ambient temperature (°C) | Charging current (A) | Discharge current (A) | Cut-off voltage (V) |
|-----------|--------------------------|----------------------|-----------------------|---------------------|
| B0005     | 24                       | 1.5                  | 2                     | 2.7                 |
| B0006     | 24                       | 1.5                  | 2                     | 2.5                 |
| B0007     | 24                       | 1.5                  | 2                     | 2.2                 |
| B0018     | 24                       | 1.5                  | 2                     | 2.5                 |
| CS2_35    | 0                        | 0.55                 | 1.1                   | 2.7                 |
| CS2_36    | 0                        | 0.55                 | 1.1                   | 2.7                 |
| CS2_37    | 0                        | 0.55                 | 1.1                   | 2.7                 |
| CS2_38    | 0                        | 0.55                 | 1.1                   | 2.7                 |
| Battery-4 | Room temperature         | 1/4/4                | 0.4/(1,2,4,6,10)/2    | 2.5/3.0/2.5         |
| Battery-5 | Room temperature         | 1/4/4                | 0.4/(1,2,4,6,10)/2    | 2.5/3.0/2.5         |
| Battery-6 | Room temperature         | 1/4/4                | 0.4/(1,2,4,6,10)/2    | 2.5/3.0/2.5         |



**Fig. 5.** The capacity and SOH degradation curves of NASA lithium-ion batteries. (a) Capacity degradation curves. (b) SOH degradation curves.

material loss hinder lithium ion diffusion, shortening the CCCT [26,27]. Charge to peak voltage time (CPVT) is the time required for the voltage to reach its peak during constant current charging. Its trend mirrors CCCT, decreasing with cycle count. Constant voltage charge time (CVCT) denotes the duration of the constant voltage charging stage. Fig. 8(b) shows the voltage profile of B0005, where CVCT prolongs with cycle count. This occurs because battery aging increases impedance, requiring more time to compensate for polarization voltage [28]. Reach peak temperature time (RPTT) is the time for the battery temperature to peak during charging. As shown in Fig. 9(a) for B0005, RPTT shortens with cycles — e.g., the peak temperature in cycle 82 is reached earlier than in cycle 42. Discharge to cut-off voltage time (DCVT) is the time for the voltage to drop to the cut-off voltage during discharging. Fig. 9(b) shows that DCVT decreases with cycle count for B0005.

The extracted HF for selected NASA, CALCE, and XJTU lithium batteries are shown in Fig. 10. Specifically, given the lack of temperature data recording in CALCE, only four HF — CCCT, CVCT, CPVT, and DCVT — are selected in this paper.

Preliminary analysis indicates significant outliers and noise in the HF. To improve model prediction accuracy and adaptability, further processing of the extracted HF is necessary.

### 2.1.3. Outlier detection

As shown in Fig. 10, significant outliers can be observed in the extracted HF. The presence of outliers may cause the model to be over-adjusted to the vicinity of the outliers, thereby reducing its overall predictive performance. Misidentifying normal data as outliers can easily destroy data integrity and reduce model effectiveness. Therefore, the correct detection of outliers is particularly important. Traditional

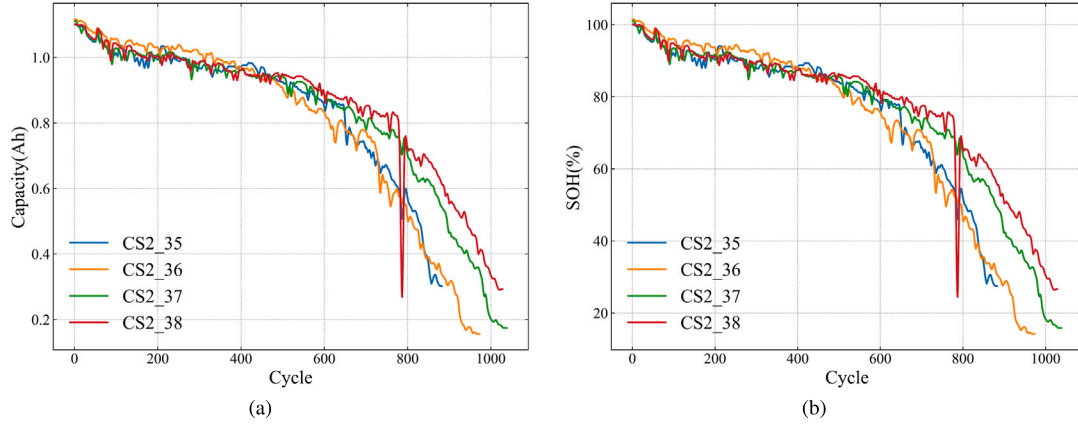


Fig. 6. The capacity and SOH degradation curves of CALCE lithium-ion batteries. (a) Capacity degradation curves. (b) SOH degradation curves.

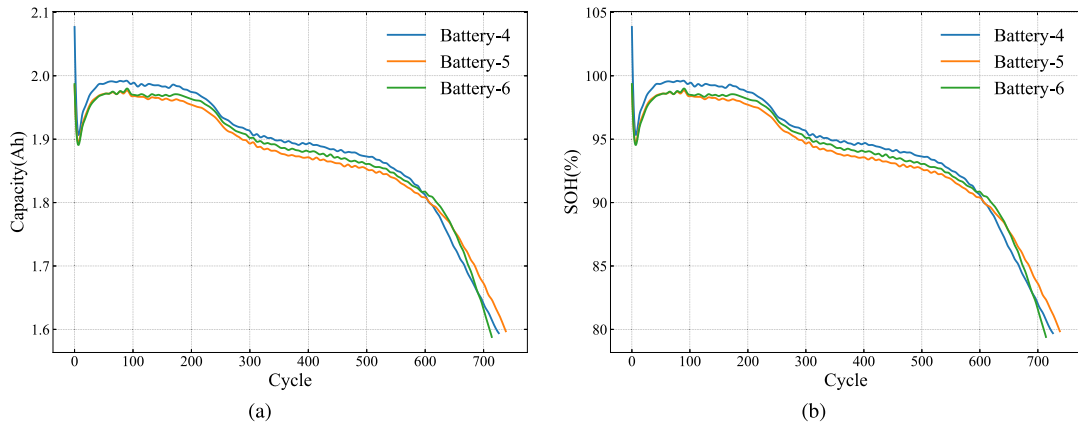


Fig. 7. The capacity and SOH degradation curves of XJTU lithium-ion batteries. (a) Capacity degradation curves. (b) SOH degradation curves.

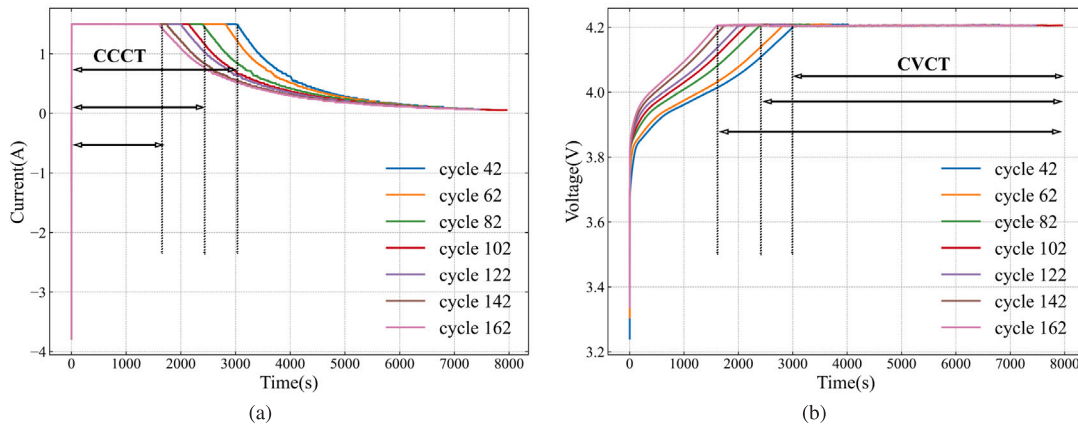


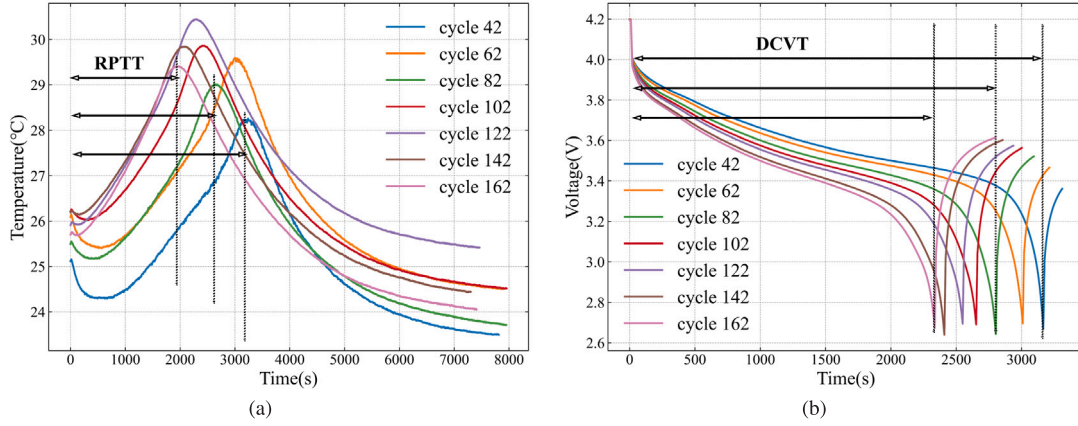
Fig. 8. Current and voltage curves during charging of lithium-ion battery B0005. (a) Current curves. (b) Voltage curves.

outlier detection methods, such as distribution-based tests and distance-based outlier detection, primarily employ a global perspective, which makes it challenging to handle complex datasets with uneven density. Normal points in high-density clusters may be misjudged as outliers, or outliers in low-density areas may be omitted. Most of them typically regard outliers as binary attributes and cannot quantify the degree of their impact. To accurately identify outliers, this paper employs the LOF algorithm to detect outliers in HF<sub>s</sub> [29]. LOF solves the limitations of traditional methods by calculating the local outlier factor of each data point and assigning a quantitative local outlier degree indicator to each

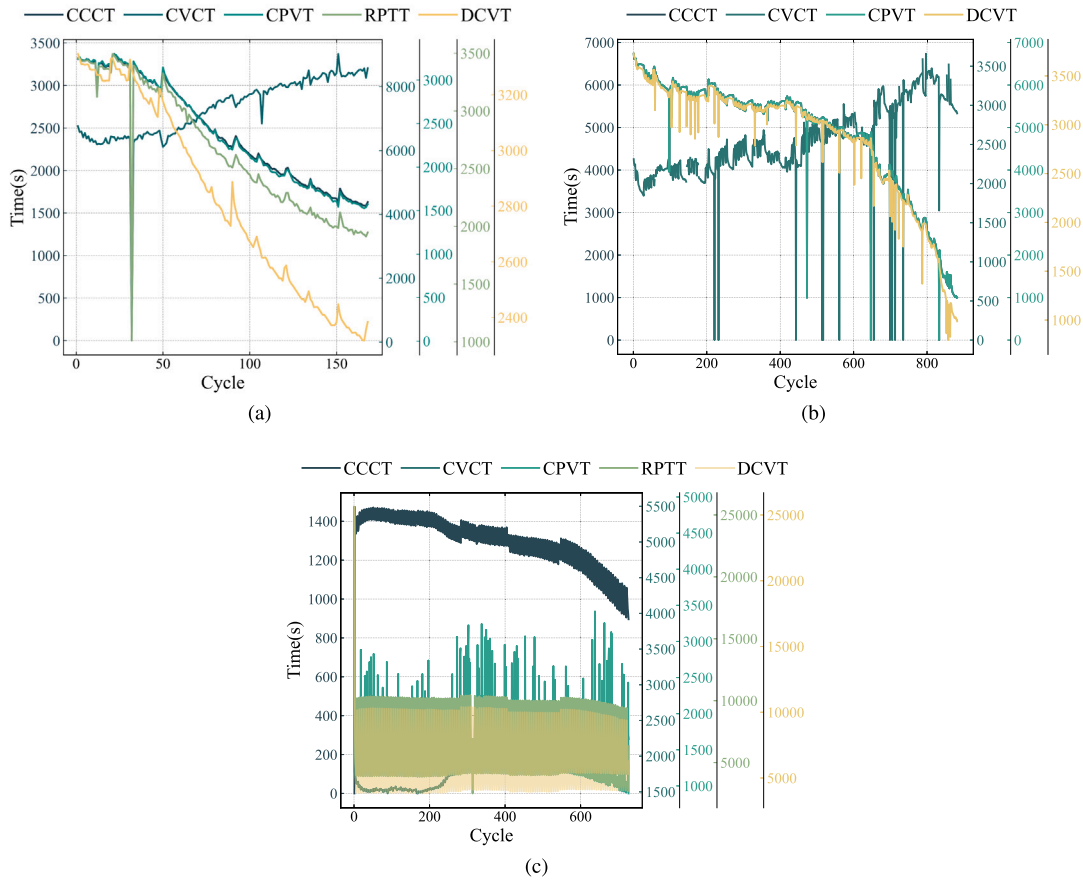
data point. Taking CCCT as an example, it is necessary to calculate the Euclidean distance of each data point in CCCT. The formula is  $D_k(i) = D_{Euclid}(i, j)$ , where  $i$  and  $j$  represent each data point. Based on the  $k$ th distance, we can calculate the  $k$ th distance neighborhood. The  $k$ th distance neighborhood is a circular area with  $i$  as the center, and the definition is shown in Eq. (2).

$$N_k(i) = \{D_{Euclid}(i, j') \leq D_k(i)\} \quad (2)$$

where  $j'$  represents the data point whose distance to  $i$  is less than or equal to  $D_k$ . The  $k$ th reachable distance is the maximum value of the



**Fig. 9.** Charging and discharging process curves of lithium-ion battery B0005. (a) Temperature curves of charging process (b) Voltage curves of discharging process.



**Fig. 10.** Curves of lithium-ion battery HFIs. (a) HFIs curves of battery B0005. (b) HFIs curves of battery CS2\_35. (c) HFIs curves of battery Battery-4.

$k$ th distance and the Euclidean distance between  $i$  and  $j$ . The reachable distance is calculated as shown in Eq. (3).

$$rd_k(i, j) = \max\{D_k(i), D_{Euclid}(i, j)\} \quad (3)$$

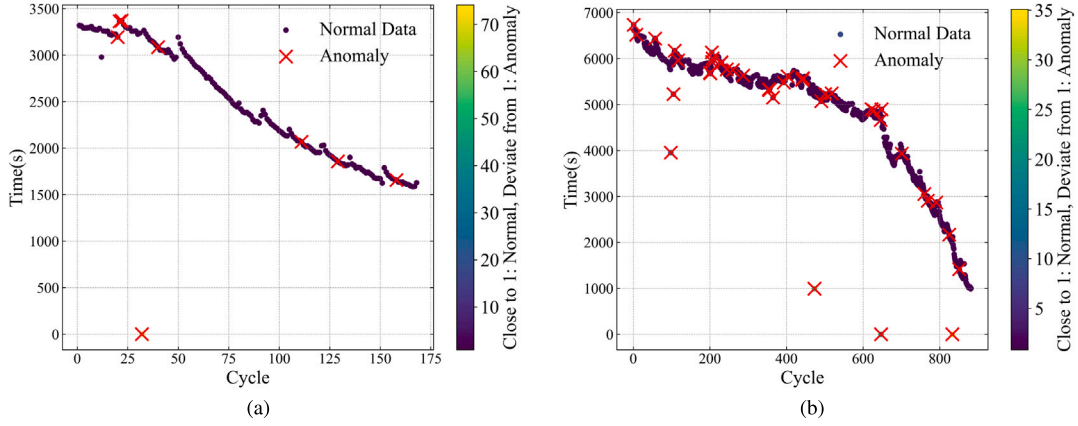
The local outlier factor is calculated based on the local reachability density (LRD). The LRD represents the reciprocal of the average  $k$ th reachable distance from all data points within the  $k$ th neighborhood of  $i$  to  $i$ . The calculation of LRD is shown in Eq. (4).

$$lrd_k(i) = \frac{N_k(i)}{\sum_{j \in N_k(i)} rd_k(i, j)} \quad (4)$$

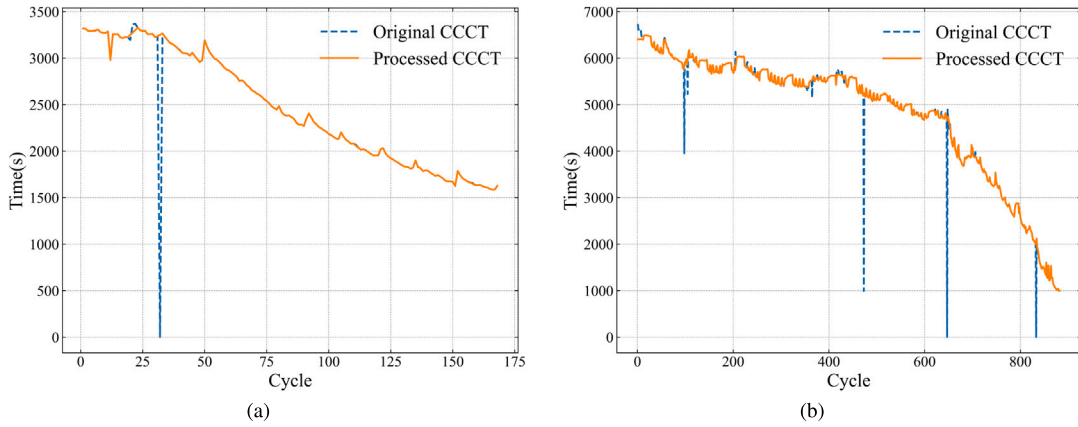
The LOF of  $i$  is the ratio of the average LRD of neighborhood data points to the LRD of  $i$ . Its calculation is shown in Eq. (5).

$$LOF_k(i) = \frac{\sum_{j \in N_k(i)} lrd_k(j)}{N_k(i) lrd_k(i)} \quad (5)$$

According to the above definitions and formulas, an LOF close to 1 indicates that the density of  $i$  is higher than that of its surrounding data points, meaning  $i$  is closer to normal conditions. Conversely, a value significantly different from 1 suggests outlier characteristics. Therefore, by calculating the LOF value for each data point in CCCT, we can detect and quantify the presence of outliers.



**Fig. 11.** The results of outlier detection for CCCT by the LOF method. (a) The outlier detection results of CCCT for battery B0005. (b) The outlier detection results of CCCT for battery CS2\_35.



**Fig. 12.** The results of outlier replacement for CCCT by Lagrange interpolation method. (a) The outlier replacement results of CCCT for battery B0005. (b) The outlier replacement results of CCCT for battery CS2\_35.

Taking the CCCT in the NASA and CALCE lithium-ion batteries as an example, outlier detection is performed on it using the LOF method. Finally, the outlier detection results of CCCT are obtained, as specifically shown in Fig. 11.

#### 2.1.4. Handling of outliers

After detecting outliers in CCCT using the LOF algorithm, directly discarding them may destroy the integrity of the data. Therefore, to ensure data integrity, Lagrange interpolation is used to replace outliers [30]. This method constructs a Lagrange polynomial through normal values adjacent to the outliers and generates smooth interpolation results to replace the outliers. The detailed steps of the Lagrange interpolation method are as follows:

**Step 1: Data Preparation.** Input the CCCT data point set where the positions of outliers have been marked by the LOF algorithm. Let the position of an outlier be  $x^*$ . Then, centered on the outlier  $x^*$ , take  $n$  normal points before and after it. These  $n$  normal points form the set  $X = \{(x_0, y_0), (x_1, y_1), \dots, (x_n, y_n)\}$ .

**Step 2: Construct Lagrange basis functions.** For each normal data point  $x_i$  in  $X$ , construct the Lagrange basis function  $L_i(x)$  as shown in Eq. (6).

$$L_i(x) = \prod_{j=0, j \neq i}^n \frac{x - x_j}{x_i - x_j} \quad (6)$$

where the basis function  $L_i(x)$  equals 1 at  $x_i$  and 0 at all other data points.

**Step 3: Generate the interpolation polynomial.** Linearly combine the basis function  $L_i(x)$  of each constructed data point  $x_i$  with the corresponding  $y_i$  to obtain the interpolation polynomial, which is calculated as shown in Eq. (7).

$$L(x) = \sum_{i=0}^n y_i \cdot L_i(x) \quad (7)$$

this polynomial passes through all data points  $x_i$  exactly.

**Step 4: Calculate the interpolation result and replace the outlier.** Substitute the position of the outlier  $x^*$  into the interpolation polynomial  $L(x)$  to calculate the estimated value  $y^* = L(x^*)$ . Finally, replace the outlier at the position  $x^*$  with the estimated value  $y^*$ .

After interpolating and replacing the outliers in CCCT using the Lagrange interpolation method, the interpolation results for the NASA and CALCE lithium-ion batteries are shown in Fig. 12.

#### 2.1.5. Noise processing

Due to the combined effects of sensor errors, environmental disturbances and data processing, noise may be introduced into the HF. Such noise can obscure real aging signals and affect the performance of the SOH prediction model. Therefore, denoising is a key step to improve data quality after outlier repair. This paper uses SSA [31] to reduce the noise in HF. SSA is a non-parametric algorithm widely applied in time-series analysis, which reduces noise in time series by extracting periodic or quasi-periodic components. Typically, SSA involves four steps: Embedding, Singular value decomposition (SVD), Grouping, and Diagonal averaging. Taking the denoising of CCCT as an example,



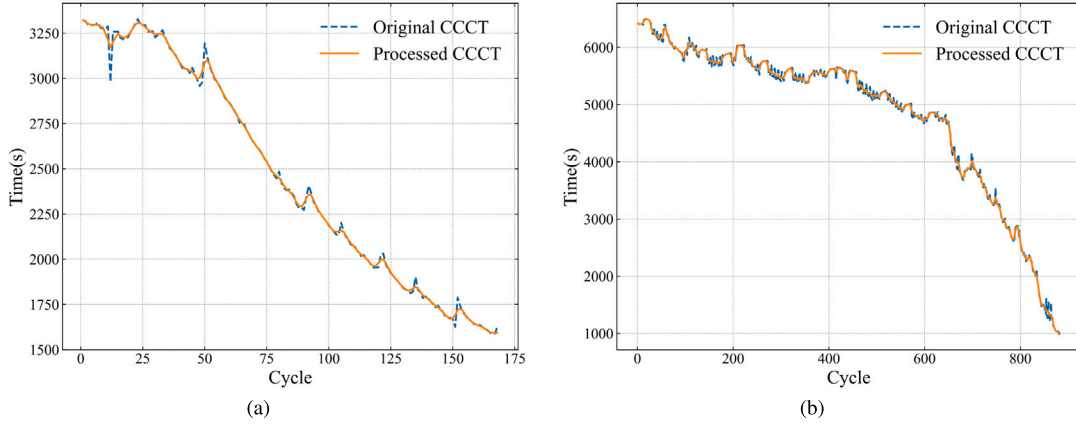


Fig. 13. The results of denoising CCCT by SSA. (a) The denoising results of CCCT for battery B0005. (b) The denoising results of CCCT for battery CS2\_35.

suppose the length of the CCCT sequence data to be denoised is  $N$ , denoted as the vector  $X = (X_1, \dots, X_N)^T$ , where the superscript  $T$  represents the transpose operator. The detailed steps of SSA denoising are as follows:

Step 1: Embedding. Convert the CCCT sequence data into a trajectory matrix  $Y = (Y_1, \dots, Y_K)$ , where  $K = N - L + 1$ ,  $L$  is the window length, and  $2 \leq L < N$ . Each element in matrix  $Y$  is defined as  $Y_i = (X_i, \dots, X_{i+L-1})^T$ , where  $i = 1, \dots, K$ , and  $Y_i$  represents an overlapping sequence obtained by dividing the CCCT time-series data with a window size of  $L$ . The matrix  $Y$  is shown in Eq. (8).

$$Y = (Y_1, \dots, Y_K) = \begin{pmatrix} X_1 & X_2 & \dots & X_K \\ X_2 & X_3 & \dots & X_{K+1} \\ \dots & \dots & \dots & \dots \\ X_L & X_{L+1} & \dots & X_N \end{pmatrix} \quad (8)$$

where the trajectory matrix  $Y$  has a dimension of  $L * K$ .

Step 2: SVD. Decompose the matrix  $Y$  into  $d$  components, where  $d$  is the rank of matrix  $Y$ . After performing singular value decomposition on matrix  $Y$  and sorting the singular values in descending order, the singular triples  $(\sqrt{\lambda_i}, U_i, V_i)$  of the matrix  $Y$  can be obtained, where  $\sqrt{\lambda_i}$ ,  $U_i$  and  $V_i$  represent the singular value, left singular vector, and right singular vector of the matrix  $Y$ , respectively.  $\lambda_i$  is the eigenvalue of the matrix  $YY^T$ . Therefore, another definition of matrix  $Y$  can be derived as shown in Eqs. (9) and (10):

$$Y = Y_1 + Y_2 + \dots + Y_d \quad (9)$$

$$Y_i = \sqrt{\lambda_i} U_i V_i^T \quad (10)$$

where  $Y_i$  are the components of matrix  $Y$ , and  $i = 1, \dots, d$ .

Step 3: Grouping. Set a threshold  $T$  (where  $0 \leq T < 1$ ) and accumulate the sorted eigenvalues  $\lambda_i$  in descending order. Denote the number of selected eigenvalues as  $m$ . When the accumulated value is greater than or equal to  $T$  times the sum of all eigenvalues, the components corresponding to these  $m$  eigenvalues are selected as the signal components of  $Y$ . Define  $I = \{I_1, \dots, I_m\}$  and  $Y_I = Y_{I_1} + \dots + Y_{I_m}$ , where  $Y_I$  represents the signal component of the CCCT sequence, and the remaining  $(d - m)$  components are treated as noise components.

Step 4: Diagonal averaging. Convert the obtained group  $\{Y_{I_1}, Y_{I_2}, \dots, Y_{I_m}\}$  into a one-dimensional time series group  $\{X_{I_1}, X_{I_2}, \dots, X_{I_m}\}$  through diagonal averaging. The original CCCT sequence data can be expressed by Eq. (11).

$$X = X_{trend} + X_{noise} = X_{I_1} + X_{I_2} + \dots + X_{I_m} + X_{noise} \quad (11)$$

After denoising CCCT using SSA, the denoised results of CCCT can be obtained from the NASA and CALCE datasets. The denoising results for NASA and CALCE lithium-ion batteries are shown in Fig. 13.

## 2.2. Dataset construction

### 2.2.1. Health factors screening

After extracting HFs from battery charge–discharge data, features characterizing battery degradation are obtained. To validate the effectiveness of the extracted HFs, this paper employs the Pearson correlation coefficient and the Spearman correlation coefficient to analyze the correlation between each HF and the SOH of lithium batteries [32,33]. The Pearson correlation coefficient, a linear measure, reflects the linear dependency between two variables, while the Spearman correlation coefficient assesses the monotonic relationship between the extracted HFs and battery SOH. The calculation formulas for the Pearson and Spearman correlation coefficients are shown in Eqs. (12) and (13), respectively.

$$PCC = \frac{\sum (X_i - \hat{X})(Y_i - \hat{Y})}{\sqrt{\sum (X_i - \hat{X})^2 \sum (Y_i - \hat{Y})^2}} \quad (12)$$

$$RHO = 1 - \frac{6 \sum_{i=1}^n d_i^2}{n(n^2 - 1)} \quad (13)$$

where  $X_i$  and  $Y_i$  represent the sample values of CCCT and battery SOH, respectively;  $\hat{X}$  and  $\hat{Y}$  are the sample means of CCCT and battery SOH;  $PCC$  is the Pearson correlation coefficient between the HF and SOH;  $d_i$  is the rank difference between  $X_i$  and  $Y_i$ , where the rank of  $X_i$  or  $Y_i$  is its position after sorting CCCT or SOH data from smallest to largest (if values are identical, their ranks are the arithmetic mean of their positions);  $RHO$  is the Spearman correlation coefficient between the HF and SOH. Both  $PCC$  and  $RHO$  range from  $-1$  to  $1$ , indicating the strength of the correlation. An absolute value between  $0.75$  and  $1$  is extremely strong,  $0.5$  to  $0.75$  is moderate, and  $0$  to  $0.25$  is weak. The closer the absolute value is to  $1$ , the stronger the association between the HF and SOH [34].

Tables 2, 3, and 4 present the correlation coefficients between HFs and SOH for the NASA, CALCE, and XJTU datasets, respectively. In this paper, an absolute correlation coefficient of  $0.5$  or higher is set as the threshold for initial screening. This threshold corresponds to a moderate level of correlation, which not only avoids missing HFs with substantial physical significance but also filters out weakly associated HFs.

The analysis shows that all HFs extracted from the NASA and CALCE datasets meet this threshold requirement — this finding further validates the effectiveness of the extracted HFs from these two datasets — while only some HFs from the XJTU dataset satisfy this requirement.

The correlation coefficient heatmap for battery B0005 is presented in Fig. 14. Analysis reveals an extremely strong correlation between RPTT and CCCT/DCVT, as well as between CPVT and CCCT/DCVT. Similar conclusions can be drawn for other batteries, as shown in Fig. 15.

**Table 2**

The Pearson correlation coefficients and Spearman correlation coefficients between different HF's and SOH of lithium-ion batteries in the NASA dataset.

| HF's | B0005    |          | B0006    |          | B0007    |          | B0018    |          |
|------|----------|----------|----------|----------|----------|----------|----------|----------|
|      | PCC      | RHO      | PCC      | RHO      | PCC      | RHO      | PCC      | RHO      |
| CCCT | 0.99720  | 0.99213  | 0.99088  | 0.99279  | 0.99428  | 0.99065  | 0.98514  | 0.96807  |
| CVCT | −0.98555 | −0.95853 | −0.92578 | −0.91497 | −0.96503 | −0.93946 | −0.53179 | −0.76209 |
| CPVT | 0.99720  | 0.99176  | 0.99067  | 0.99274  | 0.99585  | 0.99180  | 0.74283  | 0.91200  |
| RPTT | 0.99748  | 0.99354  | 0.99287  | 0.99408  | 0.99397  | 0.99013  | 0.98678  | 0.96998  |
| DCVT | 0.99961  | 0.99971  | 0.99991  | 0.99985  | 0.99896  | 0.99906  | 0.99977  | 0.99951  |

**Table 3**

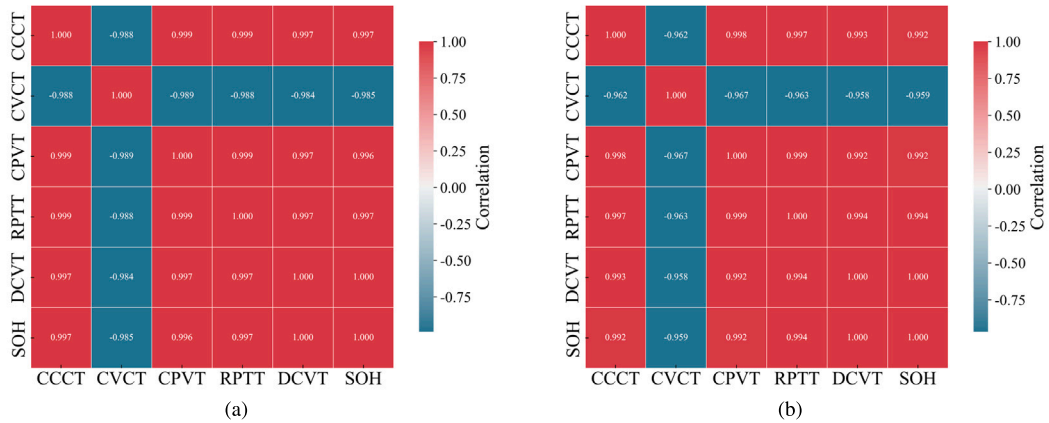
The Pearson correlation coefficients and Spearman correlation coefficients between different HF's and SOH of lithium-ion batteries in the CALCE dataset.

| HF's | CS2_35   |          | CS2_36   |          | CS2_37   |          | CS2_38   |          |
|------|----------|----------|----------|----------|----------|----------|----------|----------|
|      | PCC      | RHO      | PCC      | RHO      | PCC      | RHO      | PCC      | RHO      |
| CCCT | 0.99213  | 0.99448  | 0.99365  | 0.99581  | 0.99198  | 0.99514  | 0.97522  | 0.98053  |
| CVCT | −0.84077 | −0.90198 | −0.56925 | −0.75679 | −0.64629 | −0.83496 | −0.71595 | −0.64360 |
| CPVT | 0.99692  | 0.99470  | 0.99796  | 0.99468  | 0.99399  | 0.97749  | 0.97789  | 0.96514  |
| DCVT | 0.99817  | 0.99875  | 0.99951  | 0.99923  | 0.99911  | 0.99809  | 0.97627  | 0.99723  |

**Table 4**

The Pearson correlation coefficients and Spearman correlation coefficients between different HF's and SOH of lithium-ion batteries in the XJTU dataset.

| HF's | Battery-4 |          | Battery-5 |          | Battery-6 |          |
|------|-----------|----------|-----------|----------|-----------|----------|
|      | PCC       | RHO      | PCC       | RHO      | PCC       | RHO      |
| CCCT | 0.94525   | 0.93785  | 0.94785   | 0.94463  | 0.93261   | 0.92728  |
| CVCT | −0.94867  | −0.93444 | −0.97142  | −0.93971 | −0.95166  | −0.93381 |
| CPVT | 0.15668   | 0.62692  | 0.94728   | 0.94333  | 0.93159   | 0.92394  |
| RPTT | 0.03138   | 0.00804  | 0.01323   | −0.00146 | −0.00514  | −0.02464 |
| DCVT | 0.02281   | −0.00143 | 0.01488   | −0.00215 | −0.00229  | −0.02497 |



**Fig. 14.** The correlation coefficient matrix heatmap of lithium-ion battery B0005. (a) The Pearson correlation coefficient matrix. (b) The Spearman correlation coefficient matrix.

Therefore, to reduce feature redundancy and evaluate the cross-dataset performance of the model under the same HF's, RPTT and CPVT are excluded. Specifically, CCCT, CVCT, and DCVT are retained as input features for SOH estimation in the NASA and CALCE datasets, while only CCCT and CVCT are used as input features in the XJTU dataset, thereby constructing the experimental dataset.

### 2.2.2. Dataset partitioning

Following feature screening, the selected HF's that optimally characterize lithium-ion battery SOH degradation serve as model inputs. To address dimensional heterogeneity among HF's — which may induce computational instability or gradient explosion during training — this paper implements Z-score standardization to eliminate scale discrepancies [35]. The transformation is shown in Eq. (14).

$$z = \frac{x - \mu}{\sigma} \quad (14)$$

where  $x$  and  $z$  represent original and standardized HF's respectively, with  $\mu$  and  $\sigma$  denoting the pre-standardization mean and standard deviation.

This linear normalization generates HF's with zero mean and unit variance, accelerating model convergence. To rigorously evaluate generalization capability across battery types, an iterative partitioning scheme is adopted: For each trial, one battery's HF's from NASA, CALCE, and XJTU datasets constitute the test set, while the remaining samples are allocated as training (70%) and validation (30%) sets.

### 2.3. SOH estimation model based on BiTCN-BiGRU-MHSA

#### 2.3.1. Bidirectional temporal convolutional network model

Temporal convolutional networks (TCNs) have gained attention for improving time-series prediction [36]. Conventional CNNs are limited

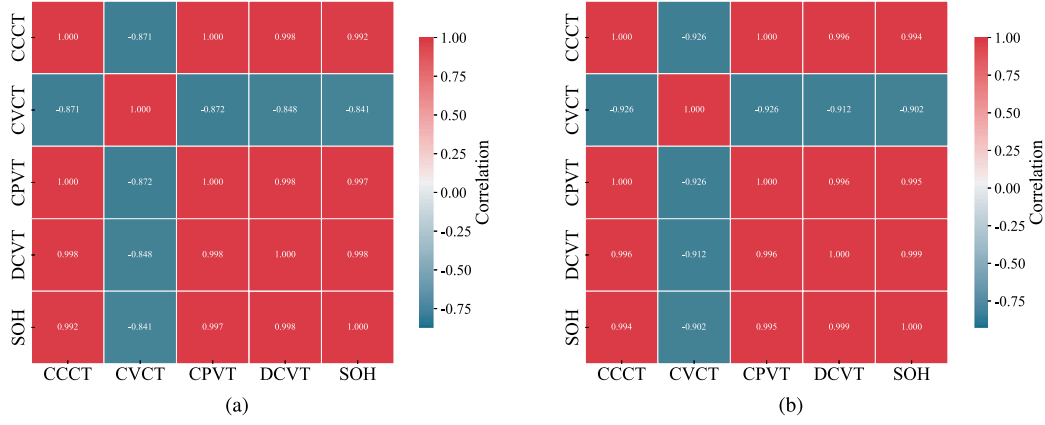


Fig. 15. The correlation coefficient matrix heatmap of lithium-ion battery CS2\_35. (a) The Pearson correlation coefficient matrix. (b) The Spearman correlation coefficient matrix.

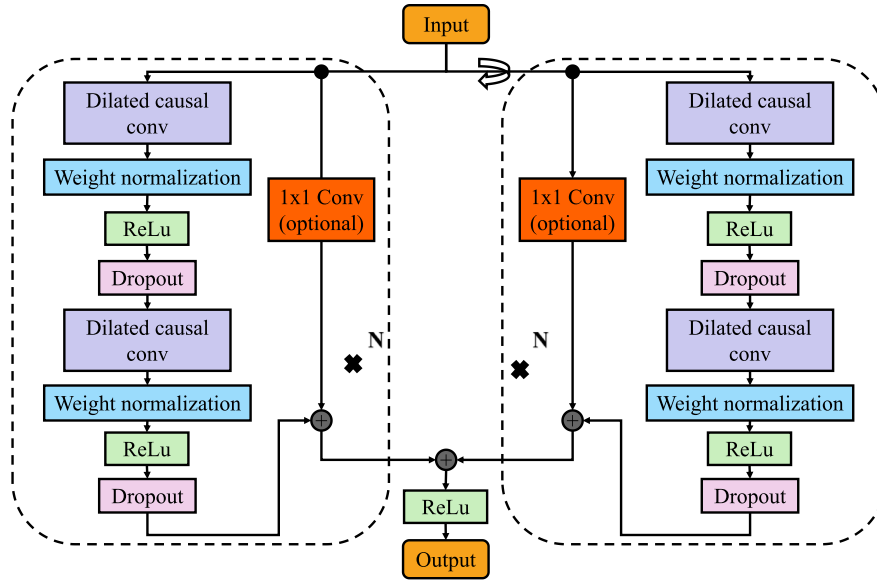


Fig. 16. The architecture of the BiTCN model.

in capturing long-term temporal dependencies by constrained kernel dimensions. TCNs mitigate this via causal and dilated convolutions, but their dilated causal convolution is unidirectional (past-to-future), failing to capture comprehensive temporal dependencies. BiTCN addresses this with parallel layers processing sequences bidirectionally (past-to-current, future-to-current), overcoming TCNs' bias [37]. Its full architecture is illustrated in Fig. 16.

Two parallel processing paths are employed in the BiTCN architecture: a forward temporal convolutional network processing sequential data chronologically to capture historical patterns, and a complementary network processing temporally reversed inputs to extract future-state information. Simultaneous integration of past/future temporal dependencies is enabled by this design. Each path has multiple stacked blocks. Dilated convolution expands the receptive field exponentially via kernel parameters, avoiding excessive depth and mitigating gradient vanishing/computational overhead. Original temporal features are preserved by residual connections to counter feature degradation. Post bidirectional extraction, forward/backward representations are fused into a holistic one encoding comprehensive temporal dynamics.

### 2.3.2. Bidirectional gated recurrent unit model

The gated recurrent unit (GRU), a variant of recurrent neural networks (RNNs) [38], is shown in Fig. 17, with two gating mechanisms:

a reset gate and an update gate. To comprehensively leverage bidirectional temporal dependencies, a BiGRU architecture is implemented, as depicted in Fig. 18. This framework processes sequential data through parallel forward and backward recurrent layers, with their respective hidden states integrated via concatenation operations. The forward GRU processes the original temporal sequence input  $X_t$ , generating the forward propagation output formalized in Eq. (15).

$$\vec{h}_t = \overline{GRU}(X_t, h_{t-1})(t = 1, 2, \dots, L) \quad (15)$$

where  $\overline{GRU}$  denotes the forward GRU computational operation, with the initial hidden state  $h_0$  initialized to zero. The backward propagation output is formalized in Eq. (16).

$$\overleftarrow{h}_t = \overline{GRU}(X_t, h_{t-1})(t = L, L-1, \dots, 1) \quad (16)$$

where  $\overline{GRU}$  represents the backward GRU operation. The concatenated hidden state output, integrating both temporal directions, is defined by Eq. (17).

$$h_t = [\vec{h}_t, \overleftarrow{h}_t] \quad (17)$$

This bidirectional integration of temporal sequences enhances feature utilization efficiency and improves predictive performance relative to conventional unidirectional GRU architectures.

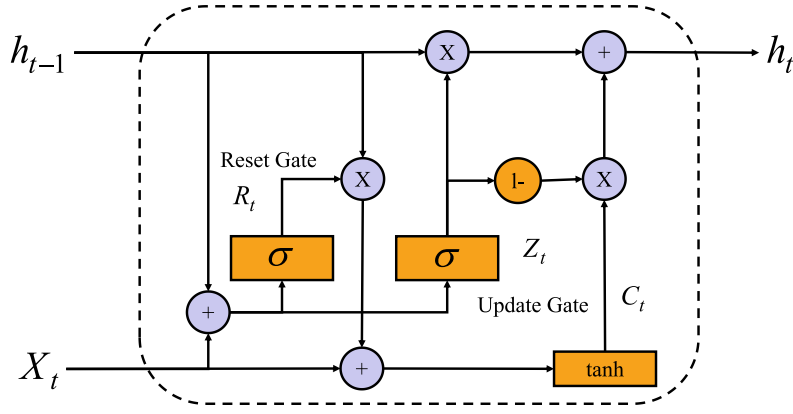


Fig. 17. The architecture of GRU.

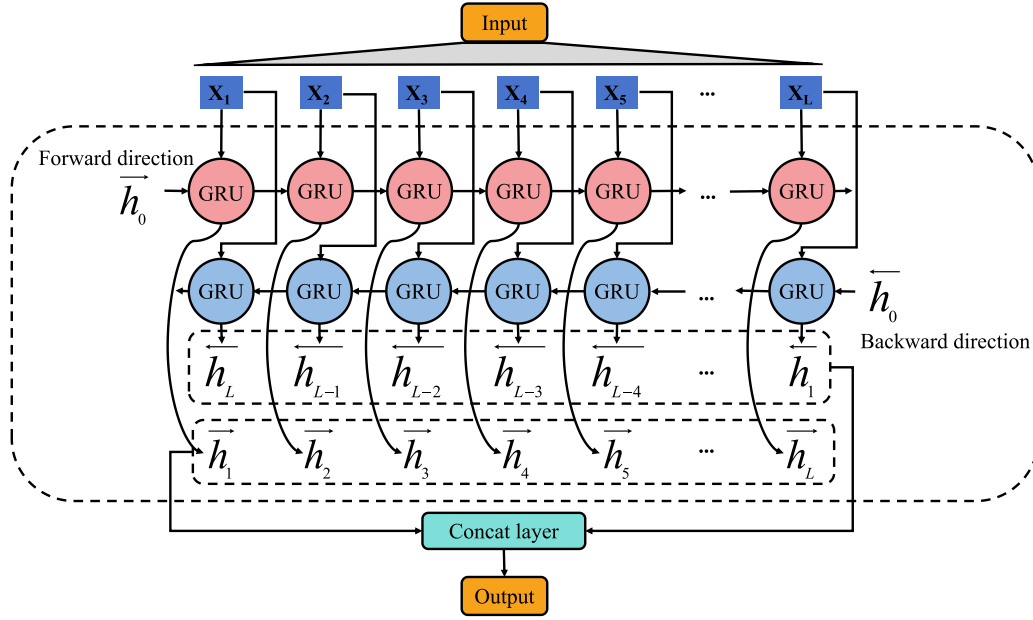


Fig. 18. The architecture of BiGRU.

### 2.3.3. Multi-head self-attention mechanism

The MHSA mechanism enables neural networks to extract salient features from input sequences  $X \in R^{n \times d_{model}}$  [39], where  $n$  denotes sequence length and  $d_{model}$  represents the feature dimension. As illustrated in Fig. 19, the input sequence undergoes linear transformations through weight matrices  $W_Q, W_K, W_V \in R^{d_{model} \times d_{model}}$  to yield query ( $Q = XW_Q$ ), key ( $K = XW_K$ ), and value ( $V = XW_V$ ) representations. These matrices are partitioned into  $h$  parallel heads along the feature dimension, generating submatrices  $Q_i, K_i, V_i \in R^{n \times d_k}$  for each head  $i = 1, 2, \dots, h$ , where  $d_{model} = h \times d_k$ . The self-attention operation for each head is calculated using Eq. (18).

$$head_i = \text{Softmax} \left( \frac{Q_i K_i^T}{\sqrt{d_k}} \right) V_i \quad (18)$$

where  $\frac{1}{\sqrt{d_k}}$  is a scaling factor to avoid gradient vanishing during Softmax normalization.

The outputs of all heads are concatenated into  $\text{Concat}(head_1, \dots, head_h) \in R^{n \times (h \times d_k)}$ , then linearly projected via weight matrix  $W_Z \in R^{(h \times d_k) \times d_{model}}$  to produce the final output  $Z = \text{Concat}(head_1, \dots, head_h) W_Z$ . This architecture captures sequential dependencies across distinct representation subspaces, significantly enhancing feature extraction capabilities.

### 2.3.4. BiTCN-BiGRU-MHSA fusion model

Conventional unidirectional architectures, including LSTM, CNN-LSTM, and GRU, exhibit limitations in modeling battery degradation dynamics. Their sequential processing constraints inhibit exploitation of future-to-current state correlations, while susceptibility to gradient vanishing or explosion phenomena impedes long-term dependency capture. To overcome these deficiencies, this paper proposes the BiTCN-BiGRU-MHSA fusion model. This architecture synergistically integrates: BiTCN's bidirectional temporal feature extraction across past and future contexts, BiGRU's dual-directional dependency modeling, and MHSA's dynamic feature weighting mechanism. As depicted in Fig. 20, BiTCN generates comprehensive feature representations through multi-scale temporal analysis. These features undergo sequential dependency encoding via BiGRU layers, while MHSA adaptively emphasizes critical degradation indicators to enhance long-term pattern recognition. Final SOH estimation is produced through a fully connected output layer, enabling robust battery health assessment.

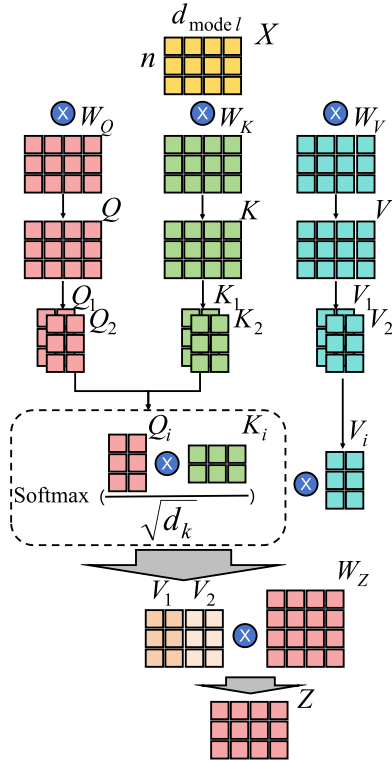
Effective hyperparameter configuration is essential for balancing predictive accuracy and generalization capability. The optimization procedure follows these steps: Initialization, Iterative training, and Model selection.

**Initialization:** Configure model parameters with learning rate, batch size, and epoch limits. Implement early stopping when validation MSE plateaus for 15 consecutive epochs.



**Table 5**  
The primary hyperparameters of the BiTCN-BiGRU-MHSA fusion model.

| Hyperparameters           | Value  |
|---------------------------|--------|
| Input dim                 | 3      |
| Output dim                | 1      |
| Input sequence length     | 16     |
| Convolutional kernel size | 3      |
| Dilation layers number    | 2      |
| Dilation rates            | [1, 2] |
| Learning rate             | 0.001  |
| Hidden size               | 80     |
| Heads number              | 5      |
| Batch size                | 32     |
| Dropout rate              | 0.1    |
| Training epoch            | 200    |
| Loss function             | MSE    |
| Optimizer                 | Adam   |



**Fig. 19.** The architecture of multi-head self-attention.

Iterative training: Iterative training encompasses the following steps.

Step 1. Propagate batched training data through the fusion model to generate SOH estimates.

Step 2. Compute batch-wise MSE loss, with its calculation method shown in Eq. (19).

$$\mathcal{L} = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2 \quad (19)$$

where  $N$  is the batch size,  $y_i$  is the actual SOH value of the  $i$ th sample,  $\hat{y}_i$  is the SOH estimation value of the  $i$ th sample output by the model, and  $\mathcal{L}$  is the MSE of the batch samples.

Step 3. Update parameters via Adam optimizer with gradient clipping.

Step 4. Terminate upon early stopping trigger or epoch exhaustion.

Model selection: Execute multiple training cycles with varied hyperparameter combinations. Select the optimal model demonstrating minimal validation MSE, then evaluate on the test set for final performance metrics.

### 3. Experiment

#### 3.1. Experiment and parameter settings

The experiments in this paper are conducted on a server equipped with the Linux operating system (version Ubuntu 22.04.1 LTS), an NVIDIA GeForce RTX 4090 D GPU, an AMD EPYC 9754 CPU, and 60 GB of memory. The proposed BiTCN-BiGRU-MHSA fusion model and comparative models are implemented using PyTorch (version 2.1.2+cu118) and Python (version 3.10.8).

For the selection of model parameters, the grid search method is used to determine hyperparameters in this paper. The Adam optimization algorithm is utilized to update the network parameters, with an initial learning rate set at 0.001. The maximum number of training epochs is fixed at 200, and the batch size is set to 32. The detailed parameter configuration of the proposed model is provided in Table 5.

To preserve comprehensive feature information, the input layer processes 3-dimensional, 16-step time-series battery data. Specifically, for the XJTU dataset, the input is 2-dimensional. In the BiTCN layer, convolutional operations employ kernels of size 3, and 2 dilation layers with dilation rates [1, 2] are utilized to progressively expand the temporal receptive field for capturing feature correlations. A dropout rate of 10% is applied in this stage to mitigate overfitting. For the BiGRU module, the hidden size is set to 80, enabling the model to capture intricate temporal dynamics of SOH variations by effectively storing historical and future dependency information in the battery degradation sequence. In the MHSA module, 5 attention heads are implemented to simultaneously extract key temporal patterns from different feature subspaces, thus enhancing the model's ability to focus on critical degradation features related to SOH.

#### 3.2. Evaluation metrics

To evaluate the performance of the proposed model, this paper employs the mean absolute error (MAE), root mean squared error (RMSE), and mean absolute percentage error (MAPE) as evaluation metrics, as shown in Eq. (20) to Eq. (22).

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (20)$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (21)$$

$$MAPE = \frac{1}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right| \times 100 \quad (22)$$

where  $y_i$  represents the actual data value,  $\hat{y}_i$  represents the model's estimated value, and  $n$  is the number of samples.

The MAE is used to measure the error between estimated values and actual values. A smaller MAE indicates a better model, and all errors are weighted equally. RMSE is a typical metric for regression

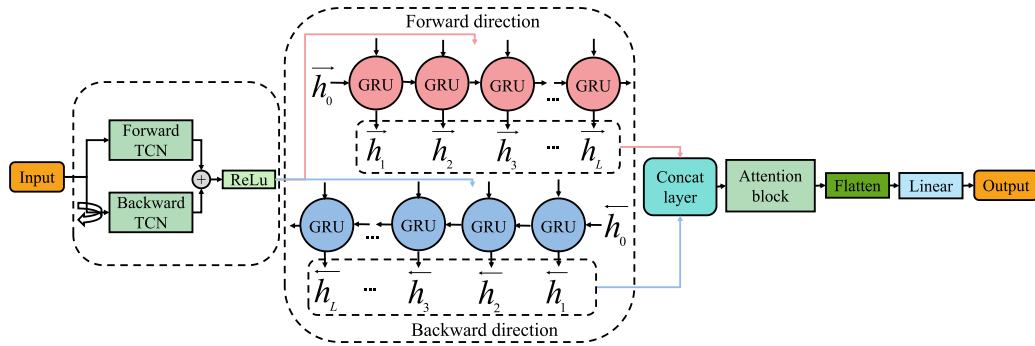


Fig. 20. The architecture of BiTCN-BiGRU-MHSA fusion model.

Table 6

The SOH estimation metrics of different models on NASA lithium-ion batteries.

| Battery | Criteria<br>(in %) | SVR  | 1D<br>CNN | LSTM | BiTCN | CNN-LSTM | CNN-CBAM-LSTM | Informer | Proposed<br>model |
|---------|--------------------|------|-----------|------|-------|----------|---------------|----------|-------------------|
| B0005   | MAE                | 0.61 | 1.06      | 1.22 | 0.78  | 0.72     | 0.72          | 1.01     | <b>0.50</b>       |
|         | RMSE               | 0.99 | 1.24      | 1.38 | 1.13  | 1.11     | 0.99          | 1.33     | <b>0.61</b>       |
|         | MAPE               | 0.83 | 1.50      | 1.73 | 1.09  | 0.98     | 1.03          | 1.43     | <b>0.69</b>       |
| B0006   | MAE                | 3.96 | 1.44      | 1.44 | 2.28  | 1.40     | 1.41          | 1.51     | <b>0.89</b>       |
|         | RMSE               | 5.64 | 1.80      | 1.98 | 2.65  | 1.86     | 2.00          | 2.04     | <b>1.33</b>       |
|         | MAPE               | 6.33 | 2.11      | 2.15 | 3.32  | 2.02     | 2.12          | 2.26     | <b>1.36</b>       |
| B0007   | MAE                | 2.72 | 1.53      | 0.89 | 1.02  | 1.00     | 0.61          | 0.87     | <b>0.49</b>       |
|         | RMSE               | 2.74 | 1.59      | 0.99 | 1.12  | 1.14     | 0.85          | 1.02     | <b>0.78</b>       |
|         | MAPE               | 3.62 | 2.03      | 1.17 | 1.35  | 1.30     | 0.79          | 1.15     | <b>0.63</b>       |
| B0018   | MAE                | 1.75 | 1.18      | 1.02 | 1.34  | 1.34     | 1.14          | 1.10     | <b>0.85</b>       |
|         | RMSE               | 1.85 | 1.41      | 1.28 | 1.59  | 1.51     | 1.33          | 1.34     | <b>0.96</b>       |
|         | MAPE               | 2.50 | 1.68      | 1.44 | 1.90  | 1.91     | 1.62          | 1.56     | <b>1.21</b>       |
| Average | MAE                | 2.26 | 1.31      | 1.14 | 1.36  | 1.11     | 0.97          | 1.12     | <b>0.68</b>       |
|         | RMSE               | 2.80 | 1.51      | 1.40 | 1.62  | 1.41     | 1.29          | 1.43     | <b>0.92</b>       |
|         | MAPE               | 3.32 | 1.83      | 1.62 | 1.91  | 1.55     | 1.39          | 1.60     | <b>0.97</b>       |

Table 7

The SOH estimation metrics of different models on CALCE lithium-ion batteries.

| Battery | Criteria<br>(in %) | SVR  | 1D<br>CNN | LSTM | BiTCN       | CNN-LSTM | CNN-CBAM-LSTM | Informer | Proposed<br>model |
|---------|--------------------|------|-----------|------|-------------|----------|---------------|----------|-------------------|
| CS2_35  | MAE                | 1.44 | 1.35      | 1.25 | 0.86        | 0.77     | 0.70          | 0.74     | <b>0.67</b>       |
|         | RMSE               | 1.73 | 1.92      | 1.93 | 1.25        | 1.05     | 1.03          | 1.03     | <b>0.84</b>       |
|         | MAPE               | 4.74 | 4.34      | 4.29 | 3.04        | 2.55     | 2.53          | 2.57     | <b>1.99</b>       |
| CS2_36  | MAE                | 1.11 | 1.47      | 1.09 | 1.26        | 1.02     | 0.77          | 1.05     | <b>0.61</b>       |
|         | RMSE               | 3.54 | 3.01      | 1.85 | 1.77        | 1.81     | 0.99          | 1.74     | <b>0.85</b>       |
|         | MAPE               | 9.28 | 10.77     | 5.99 | 6.70        | 5.75     | 3.91          | 7.13     | <b>2.77</b>       |
| CS2_37  | MAE                | 1.38 | 0.80      | 0.96 | 0.87        | 1.01     | 0.70          | 0.73     | <b>0.34</b>       |
|         | RMSE               | 1.60 | 1.23      | 1.22 | 1.02        | 1.18     | 0.97          | 0.85     | <b>0.48</b>       |
|         | MAPE               | 6.31 | 3.85      | 4.66 | 3.28        | 3.78     | 3.97          | 3.14     | <b>1.45</b>       |
| CS2_38  | MAE                | 1.63 | 1.40      | 1.50 | 1.34        | 1.46     | 1.33          | 1.36     | <b>1.05</b>       |
|         | RMSE               | 3.88 | 3.77      | 3.84 | <b>3.66</b> | 3.74     | 3.70          | 3.76     | <b>3.68</b>       |
|         | MAPE               | 7.22 | 6.41      | 6.74 | 6.31        | 6.55     | 6.12          | 6.11     | <b>5.00</b>       |
| Average | MAE                | 1.39 | 1.25      | 1.20 | 1.08        | 1.07     | 0.87          | 0.97     | <b>0.67</b>       |
|         | RMSE               | 2.69 | 2.49      | 2.21 | 1.93        | 1.95     | 1.67          | 1.84     | <b>1.46</b>       |
|         | MAPE               | 6.89 | 6.34      | 5.42 | 4.83        | 4.65     | 4.13          | 4.74     | <b>2.80</b>       |

Table 8

The SOH estimation metrics of different models on XJTU lithium-ion batteries.

| Battery   | Criteria<br>(in %) | SVR   | 1D CNN | LSTM  | BiTCN | CNN-LSTM | CNN-CBAM-LSTM | Informer     | Proposed<br>model |
|-----------|--------------------|-------|--------|-------|-------|----------|---------------|--------------|-------------------|
| Battery-4 | MAE                | 0.840 | 0.154  | 0.135 | 0.100 | 0.119    | 0.072         | 0.081        | <b>0.052</b>      |
|           | RMSE               | 0.928 | 0.172  | 0.158 | 0.124 | 0.142    | 0.091         | 0.125        | <b>0.070</b>      |
|           | MAPE               | 0.905 | 0.170  | 0.147 | 0.108 | 0.129    | 0.078         | 0.087        | <b>0.057</b>      |
| Battery-5 | MAE                | 0.510 | 0.138  | 0.117 | 0.109 | 0.085    | <b>0.069</b>  | 0.088        | 0.081             |
|           | RMSE               | 0.589 | 0.153  | 0.135 | 0.132 | 0.109    | <b>0.089</b>  | 0.128        | 0.110             |
|           | MAPE               | 0.562 | 0.152  | 0.128 | 0.119 | 0.093    | <b>0.075</b>  | 0.096        | 0.089             |
| Battery-6 | MAE                | 0.193 | 0.147  | 0.124 | 0.108 | 0.096    | 0.107         | <b>0.091</b> | 0.093             |
|           | RMSE               | 0.265 | 0.172  | 0.146 | 0.134 | 0.132    | <b>0.132</b>  | <b>0.132</b> | 0.136             |
|           | MAPE               | 0.212 | 0.164  | 0.135 | 0.117 | 0.104    | 0.118         | <b>0.098</b> | 0.100             |
| Average   | MAE                | 0.514 | 0.146  | 0.125 | 0.106 | 0.100    | 0.083         | 0.087        | <b>0.075</b>      |
|           | RMSE               | 0.594 | 0.166  | 0.146 | 0.130 | 0.128    | <b>0.104</b>  | 0.128        | 0.105             |
|           | MAPE               | 0.560 | 0.162  | 0.137 | 0.115 | 0.109    | 0.090         | 0.094        | <b>0.082</b>      |

**Table 9**  
The SOH estimation metrics of transformer-based models.

| Battery   | Criteria (in %) | Transformer  | FEDFormer    | Proposed model |
|-----------|-----------------|--------------|--------------|----------------|
| B0005     | MAE             | 0.548        | <b>0.429</b> | 0.504          |
|           | RMSE            | 0.763        | 0.758        | <b>0.605</b>   |
|           | MAPE            | 0.761        | <b>0.597</b> | 0.690          |
| B0006     | MAE             | 0.936        | <b>0.846</b> | 0.893          |
|           | RMSE            | 1.345        | 1.620        | <b>1.326</b>   |
|           | MAPE            | 1.360        | <b>1.195</b> | 1.357          |
| B0007     | MAE             | 0.577        | 0.530        | <b>0.487</b>   |
|           | RMSE            | <b>0.758</b> | 0.834        | 0.775          |
|           | MAPE            | 0.754        | 0.701        | <b>0.630</b>   |
| B0018     | MAE             | 1.064        | 1.027        | <b>0.848</b>   |
|           | RMSE            | 1.603        | 1.611        | <b>0.958</b>   |
|           | MAPE            | 1.500        | 1.450        | <b>1.207</b>   |
| CS2_35    | MAE             | 0.746        | 0.752        | <b>0.670</b>   |
|           | RMSE            | 1.038        | 1.200        | <b>0.840</b>   |
|           | MAPE            | 2.757        | 2.650        | <b>1.990</b>   |
| CS2_36    | MAE             | 0.742        | 0.750        | <b>0.610</b>   |
|           | RMSE            | 1.028        | 1.144        | <b>0.850</b>   |
|           | MAPE            | 3.108        | 3.589        | <b>2.774</b>   |
| CS2_37    | MAE             | 0.493        | 0.392        | <b>0.338</b>   |
|           | RMSE            | 0.642        | 0.545        | <b>0.477</b>   |
|           | MAPE            | 1.972        | <b>1.416</b> | 1.450          |
| CS2_38    | MAE             | 1.534        | 1.111        | <b>1.048</b>   |
|           | RMSE            | 3.771        | 3.716        | <b>3.682</b>   |
|           | MAPE            | 5.613        | 5.033        | <b>5.001</b>   |
| Battery-4 | MAE             | 0.114        | 0.066        | <b>0.052</b>   |
|           | RMSE            | 0.144        | 0.080        | <b>0.070</b>   |
|           | MAPE            | 0.122        | 0.072        | <b>0.057</b>   |
| Battery-5 | MAE             | 0.181        | 0.100        | <b>0.081</b>   |
|           | RMSE            | 0.250        | 0.122        | <b>0.110</b>   |
|           | MAPE            | 0.208        | 0.112        | <b>0.089</b>   |
| Battery-6 | MAE             | 0.207        | 0.100        | <b>0.093</b>   |
|           | RMSE            | 0.318        | <b>0.127</b> | 0.136          |
|           | MAPE            | 0.238        | 0.111        | <b>0.100</b>   |
| Average   | MAE             | 0.649        | 0.555        | <b>0.511</b>   |
|           | RMSE            | 1.060        | 1.069        | <b>0.894</b>   |
|           | MAPE            | 1.672        | 1.539        | <b>1.395</b>   |

models, used to indicate the magnitude of errors produced by the model in predictions. For larger errors, RMSE places higher weights. MAPE is a metric employed to evaluate the accuracy of predictive models. It reflects the average percentage error between estimated values and actual values, measuring the proportion of the difference between estimated and actual values relative to the actual values. The larger the error, the greater the MAPE.

### 3.3. Comparative experiments

The efficacy of the proposed fusion model is rigorously validated through comparative analysis against established models, with experimental results visualized for NASA, CALCE, and XJTU datasets in Figs. 21, 22, and 23. While comparative models generally capture macroscopic SOH degradation trends, significant accuracy disparities emerge under quantitative scrutiny. As detailed in Tables 6, 7, and 8, the proposed model demonstrates superior performance across RMSE, MAE, and MAPE metrics on both datasets.

Traditional machine learning approaches exhibit notable limitations; the SVR model shows substantial performance variance across batteries, particularly evident in NASA B0006 (MAE: 3.96%, RMSE: 5.64%). This instability stems from SVR's inability to model long-term temporal dependencies and its empirical dependence on kernel selection.

Among single-architecture deep learning models, the 1D-CNN captures local temporal features but fails to address long-term dependencies, yielding a NASA MAE of 1.31%. While LSTM networks offer enhanced temporal modeling, their standalone implementation still underperforms due to limited multi-scale feature extraction.

Fusion architectures demonstrate incremental improvements yet retain fundamental constraints. The BiTCN model enhances bidirectional perception but remains inferior to both Informer and the proposed method in long-term dependency capture. CNN-LSTM achieves

marginal accuracy gains over LSTM through convolutional feature extraction but remains hampered by unidirectional processing. The CNN-CBAM-LSTM variant reduces average MAE by 13% (NASA), 19% (CALCE), and 17%(XJTU) relative to CNN-LSTM through attention mechanisms [40], though overall performance remains constrained.

The Informer model, leveraging ProbSparse self-attention [41], significantly advances long-term dependency modeling with reduced computational complexity. It outperforms most conventional and deep learning models but exhibits higher errors than the proposed method due to deficient multi-scale local feature extraction.

The performance of the proposed model is further validated by comparing it with two advanced transformer-based models using the metrics in Table 9. On the NASA dataset, for instance, the MAE of B0007 is reduced by 16% compared to Transformer and 8% compared to FEDFormer [42], while the MAE of B0018 is reduced by over 17% relative to both models. On the CALCE dataset, the MAE of CS2\_37 is decreased by 31% versus Transformer and 14% versus FEDFormer. Even on the XJTU validation set, the MAE of Battery-4 is 21% lower than that of FEDFormer. As an improved version of the Transformer, FEDFormer effectively makes up for the Transformer's shortcomings in computational efficiency and fine-grained feature capture, achieving comprehensive performance improvements in SOH estimation while maintaining the Transformer's advantage in long-term modeling. However, both models still fail to model bidirectional temporal correlations.

Ultimately, the proposed method achieves lower errors on NASA, CALCE, and XJTU datasets, with more pronounced gains on NASA data (30% and 39% reductions in MAE versus CNN-CBAM-LSTM and Informer, respectively). These results conclusively validate the model's capacity to synergistically integrate bidirectional temporal processing with attention-weighted feature selection for superior SOH estimation.

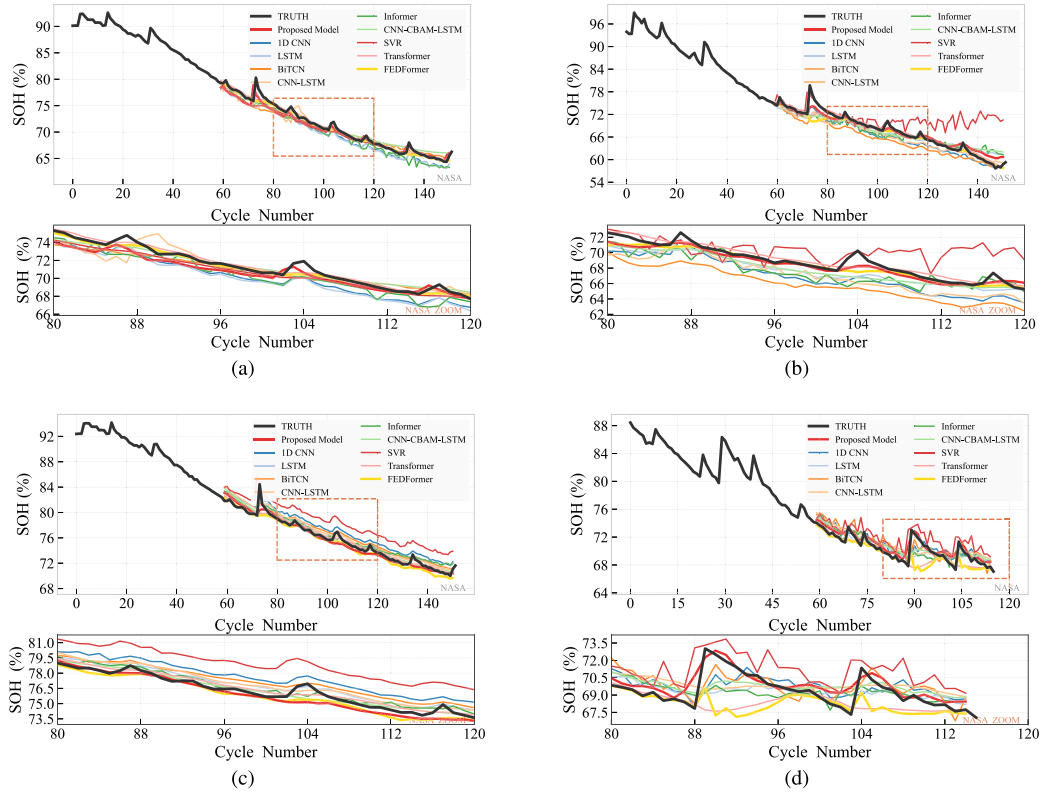


Fig. 21. The comparative experimental results of NASA lithium-ion batteries. (a) B0005. (b) B0006. (c) B0007. (d) B0018.

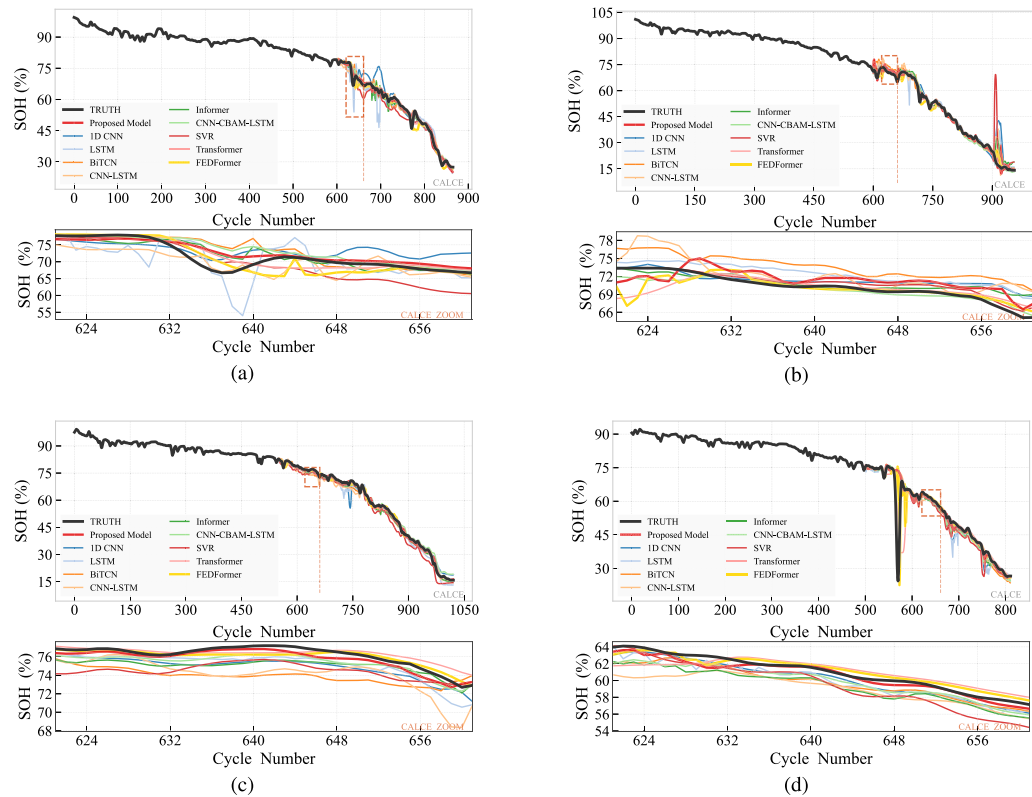


Fig. 22. The comparative experimental results of CALCE lithium-ion batteries. (a) CS2\_35. (b) CS2\_36. (c) CS2\_37. (d) CS2\_38.

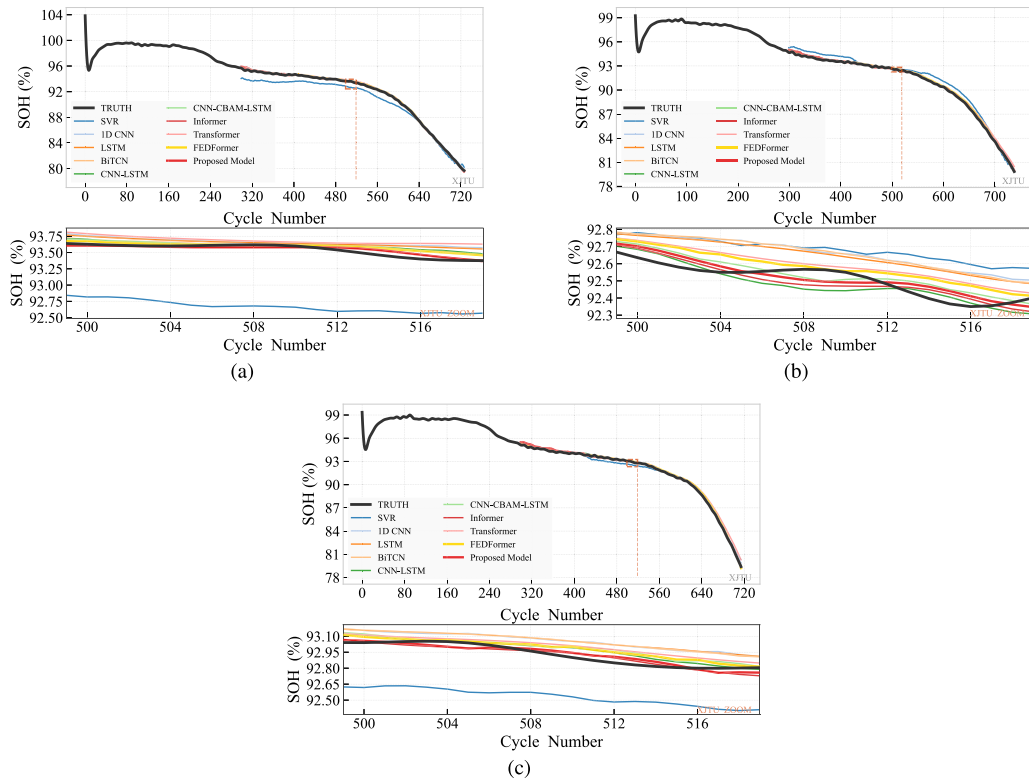


Fig. 23. The comparative experimental results of XJTU lithium-ion batteries. (a) Battery-4. (b) Battery-5. (c) Battery-6.

Table 10

The results of ablation experiment.

| Model            | Module |       |      | Average MAE (in %) |             |             |             |             |
|------------------|--------|-------|------|--------------------|-------------|-------------|-------------|-------------|
|                  | BiTCN  | BiGRU | MHSA | 0–25               | 25–50       | 50–75       | 75–100      | 0–100       |
| BiTCN-BiGRU-MHSA | ✓      | ✓     | ✓    | <b>1.02</b>        | <b>1.17</b> | <b>0.57</b> | <b>0.68</b> | <b>0.85</b> |
| BiTCN-BiGRU      | ✓      | ✓     |      | 1.47(+44%)         | 1.55(+33%)  | 1.44(+153%) | 1.14(+68%)  | 1.40(+65%)  |
| BiTCN-MHSA       | ✓      |       | ✓    | 1.20(+18%)         | 1.70(+45%)  | 1.19(+109%) | 1.82(+168%) | 1.48(+74%)  |
| BiGRU-MHSA       |        | ✓     | ✓    | 1.25(+23%)         | 2.11(+80%)  | 1.36(+139%) | 1.39(+104%) | 1.53(+80%)  |
| TCN-BiGRU-MHSA   |        | ✓     | ✓    | 1.64(+61%)         | 1.45(+24%)  | 1.13(+98%)  | 2.15(+216%) | 1.59(+87%)  |
| BiTCN-GRU-MHSA   | ✓      |       | ✓    | 2.33(+128%)        | 1.63(+39%)  | 1.37(+140%) | 1.15(+69%)  | 1.62(+91%)  |

### 3.4. Ablation experiment

To validate synergistic effects among modules in the proposed fusion model, a systematic ablation study is conducted by removing or replacing core modules. Performance is evaluated using MAE across degradation stages (0%–25%, 25%–50%, 50%–75%, 75%–100%) for batteries B0005, B0006, B0007 and B0018, with results summarized in Table 10 and visualizations presented in Fig. 24

Experimental findings demonstrate the model's significant advantage throughout the full degradation cycle (0%–100%), achieving an overall MAE of 0.85%—representing a 39% reduction compared to the optimal baseline. This superiority is particularly pronounced during the terminal degradation phase (75%–100%), where MAE decreases to 0.68%. Compared to the BiTCN-BiGRU model without MHSA and the unidirectional TCN-BiGRU-MHSA variant, errors are reduced by 40% and 68%, respectively, confirming the necessity of bidirectional feature extraction integrated with multi-head self-attention.

Module-specific contributions are rigorously characterized across degradation phases. During early degradation (0%–25%), the BiTCN module maintains MAE at 1.02%, significantly outperforming the BiGRU-MHSA model lacking this module (MAE = 1.25%, a 23% increase in MAE). In mid-term degradation (25%–50%), the BiGRU

module effectively models temporal dependencies. Bidirectional gating mechanisms sustain MAE at 1.17%, substantially lower than the BiTCN-MHSA model missing this module (MAE = 1.70%, a 45% increase in MAE). During terminal degradation (75%–100%), the MHSA module proves critical. The complete model achieves MAE = 0.68%, while the MHSA-ablated variant degrades to MAE = 1.14%. Notably, the bidirectional architecture is indispensable: Replacing the bidirectional TCN with a unidirectional structure (TCN-BiGRU-MHSA) causes a 216.2% increase in MAE during the terminal degradation phase, while downgrading the bidirectional GRU to unidirectional (BiTCN-GRU-MHSA) increases early degradation MAE by 128.4%.

Collectively, these results confirm that integrating BiTCN, BiGRU, and MHSA significantly enhances SOH estimation performance while validating each module's efficacy.

### 3.5. Validation on the effectiveness and validity of health factors

To validate the effectiveness and validity of HFs, different combinations of HFs are designed, and experiments are further conducted on the XJTU dataset, respectively, with the combination information presented in Table 11 and the experimental results shown in Table 12.

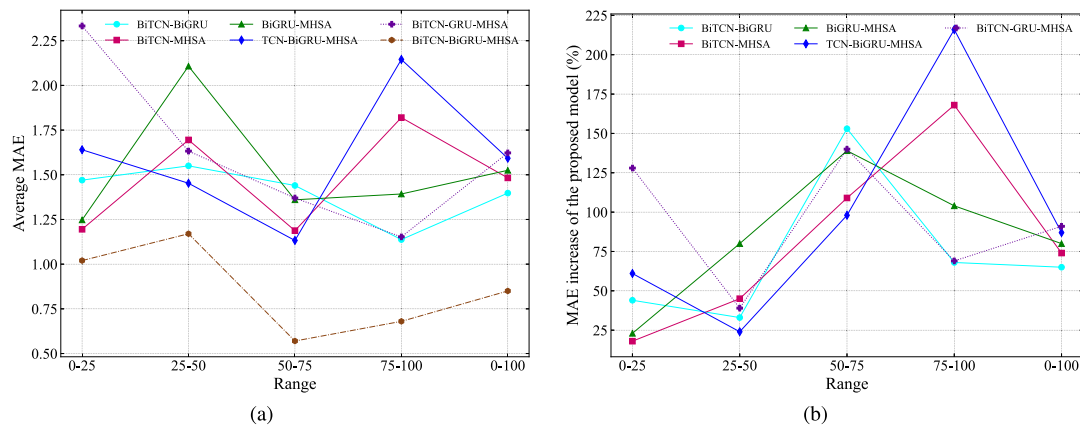


**Table 11**  
The information on HF combinations.

| Combination                  | Abbreviation       |
|------------------------------|--------------------|
| CCCT                         | HF <sub>s</sub> -1 |
| CCCT, CVCT, DCVT             | HF <sub>s</sub> -2 |
| CCCT, CVCT, DCVT, RPTT, CPVT | HF <sub>s</sub> -3 |
| CCCT, DCVT                   | HF <sub>s</sub> -4 |
| CVCT                         | HF <sub>s</sub> -5 |
| CVCT, DCVT                   | HF <sub>s</sub> -6 |
| DCVT                         | HF <sub>s</sub> -7 |
| CCCT, CVCT                   | Proposed           |

**Table 12**  
The validation results of the effectiveness and validity of HF<sub>s</sub>.

| Battery   | Criteria (in %) | HF <sub>s</sub> -1 | HF <sub>s</sub> -2 | HF <sub>s</sub> -3 | HF <sub>s</sub> -4 | HF <sub>s</sub> -5 | HF <sub>s</sub> -6 | HF <sub>s</sub> -7 | Proposed     |
|-----------|-----------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------|
| Battery-4 | MAE             | 0.086              | 0.064              | 0.112              | 0.070              | 0.105              | 0.080              | 0.160              | <b>0.052</b> |
|           | RMSE            | 0.129              | 0.093              | 0.184              | 0.108              | 0.124              | 0.100              | 0.188              | <b>0.070</b> |
|           | MAPE            | 0.093              | 0.070              | 0.128              | 0.076              | 0.116              | 0.087              | 0.173              | <b>0.057</b> |
| Battery-5 | MAE             | 0.089              | 0.085              | 0.086              | 0.098              | 0.129              | 0.100              | 0.125              | <b>0.081</b> |
|           | RMSE            | 0.127              | 0.126              | 0.124              | 0.133              | 0.151              | 0.120              | 0.146              | <b>0.110</b> |
|           | MAPE            | 0.096              | 0.093              | 0.094              | 0.108              | 0.141              | 0.109              | 0.137              | <b>0.089</b> |
| Battery-6 | MAE             | 0.105              | 0.087              | 0.106              | <b>0.085</b>       | 0.103              | 0.107              | 0.193              | 0.093        |
|           | RMSE            | 0.141              | 0.129              | 0.149              | <b>0.123</b>       | 0.128              | 0.125              | 0.214              | 0.136        |
|           | MAPE            | 0.115              | 0.094              | 0.116              | <b>0.092</b>       | 0.112              | 0.116              | 0.214              | 0.100        |
| Average   | MAE             | 0.093              | 0.079              | 0.101              | 0.084              | 0.112              | 0.096              | 0.159              | <b>0.075</b> |
|           | RMSE            | 0.132              | 0.116              | 0.152              | 0.121              | 0.134              | 0.115              | 0.183              | <b>0.105</b> |
|           | MAPE            | 0.101              | 0.086              | 0.113              | 0.092              | 0.123              | 0.104              | 0.175              | <b>0.082</b> |



**Fig. 24.** The results of ablation experiment. (a) MAE. (b) Relative to the proposed model, the MAE change of other models.

The experimental results show that the proposed HF<sub>s</sub> achieved the optimal performance across all batteries and the average metrics, verifying the superiority of the proposed HF<sub>s</sub> combination.

#### 4. Conclusions

Accurate estimation of the SOH of lithium-ion batteries is crucial for ensuring the reliable management and maintenance of battery systems in critical applications such as new energy vehicles and aerospace. Most existing data-driven SOH estimation methods have challenges: they struggle to fully exploit bidirectional temporal correlations in battery degradation trajectories and insufficiently capture long-term aging dependencies, which restricts estimation accuracy. In this paper, a comprehensive framework for SOH estimation is proposed, including an HF processing pipeline and a novel BiTCN-BiGRU-MHSA fusion model. The framework integrates HF extraction, outlier mitigation, denoising, feature screening, and deep learning modeling, with validation conducted on NASA, CALCE, and XJTU lithium-ion battery datasets. Detailed research status analysis, method descriptions, implementation processes, and comparative result analyses are provided, leading to the following conclusions:

(1) The proposed HF extraction and preprocessing pipeline improves data quality. HF<sub>s</sub> indicative of SOH degradation are extracted through a comprehensive analysis of current, voltage, and temperature profiles from battery datasets. These HF<sub>s</sub> undergo preprocessing through a pipeline combining LOF detection with Lagrange interpolation for outlier replacement, integrated with SSA denoising to preserve authentic aging signatures while enhancing data fidelity. Subsequent feature selection employs a combined Pearson-Spearman correlation analysis to identify degradation-sensitive inputs, resulting in the effective elimination of redundant features and optimization of dataset quality.

(2) The BiTCN-BiGRU-MHSA fusion model addresses the issues of traditional methods by integrating bidirectional feature extraction, long-term dependency modeling, and feature weighting. The BiTCN captures bidirectional temporal correlations, overcoming the one-way bias of conventional architectures; the BiGRU models long-term aging dependencies across time steps; the MHSA mechanism emphasizes critical degradation patterns. The experimental results indicate that the model achieves average MAE of 0.68% (NASA), 0.67% (CALCE), and 0.075% (XJTU), representing reductions of 30%/23%/10% compared to the CNN-CBAM-LSTM model and 39%/31%/14% compared to the Informer model, respectively.

Notwithstanding its demonstrated efficacy, the proposed fusion model incurs elevated computational complexity when processing long sequences due to bidirectional operations and attention mechanisms. This limitation prolongs training durations and may constrain deployment in resource-limited scenarios. Future research should prioritize lightweight architectural optimizations to enhance operational efficiency.

### CRedit authorship contribution statement

**Qingzhong Tang:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Ting He:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition. **Jie Huang:** Writing – review & editing, Formal analysis. **Wenlong Zhu:** Writing – review & editing, Formal analysis.

### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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### Data availability

The authors do not have permission to share data.

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